

Unified Agent Benchmark (HIL-Bench): Canonical Testing Methods for Unified Intelligence and the Harness Intelligence Level

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Abstract

The companion paper, *Unified Intelligence Theory and the Harness Intelligence Level*, defines a cumulative intelligence hierarchy built from Cognitive Intelligence C , Individual Intelligence I , Organizational Intelligence O , Delegation Intelligence DI , and operational Self-Awareness SA , with directly measured coordinates $[C, I, O, T, H, SA]$. It preserves the gated Unified Intelligence hierarchy $U0-U\Omega$ and clarifies Individual Intelligence as the longitudinal evolution of one persistent decision locus: transient operation ($I0$), persistent continuity ($I1$), adaptive learning under a fixed learning mechanism ($I2$), governed self-improvement ($I3$), recursive improvement of improvement ($I4$), persistent autonomous discovery ($I5$), and open-ended expansion of future cognitive reach ($I\Omega$). The theory further proposes an invariance principle: after a standard is ratified, models should move through the scale; the meaning of the scale should not move with the models.

This paper defines the corresponding **measurement standard**. HIL-Bench separates an invariant **canonical test law** from the concrete task instances used to obtain evidence. For each coordinate level, the standard specifies the experimental intervention, allowed resources, human-intervention budget, external verifier, lower-level retention requirements, reliability criterion, and admissible evidence. A fixed public reference set makes every level inspectable and reproducible. A fixed private anchor set supplies a stable historical certification reference. New private witness forms may be added for contamination resistance only when they implement the same canonical test law; they extend evidence without redefining the level. Some upper-level claims use frozen human-review protocols when objective automatic verification is insufficient, but the rubric, blinding, admissibility criteria, and acceptance rule are part of the standard rather than discretionary judgment.

HIL-Bench evaluates two subjects. In **agent mode**, a frozen model-harness pair is measured directly and receives a gated Unified level, the continuous pair score $HLIS$, the coordinate profile $[C, I, O, T/H, SA]$, delegation frontiers, false-completion rate, uncertainty, and a declared resource envelope. In **model mode**, one frozen base LLM is evaluated through a standardized reference harness ladder, yielding a Harness Scaling Curve, HIL-Level, HIL-AUC, HIL-Ceiling, Harness Gain, verification responsiveness ρ_V , and an optional composite HIL-Score. The curve and gated level remain primary.

The main contribution is therefore not another moving benchmark corpus. It is a candidate **stable experimental constitution for intelligence measurement**: level semantics are defined in Paper 1; canonical experiments are defined here; datasets are witnesses to those experiments. Benchmark saturation motivates higher levels or additional witness forms, not movement of an existing level boundary.

Keywords: Harness Intelligence Level; HIL-Bench; Unified Intelligence; agent benchmark; LLM benchmark; persistence; adaptive learning; self-improvement; recursive improvement; discovery; public-private evaluation; canonical testing

1. Introduction

Most AI benchmarks are deliberately narrow. Multiple-choice science benchmarks measure answer accuracy; coding benchmarks execute programs against tests; repository benchmarks verify patches; interactive benchmarks measure

behavior in an environment. These instruments are useful precisely because their scoring rules are concrete. They become less informative when a single result is interpreted as a complete intelligence level.

The companion theory takes the opposite direction: it defines a structured intelligence hierarchy and asks what evidence is required to support each claim. That creates a measurement problem. If the test changes every time frontier models improve, then an intelligence level becomes a moving percentile rather than a stable scientific concept. If the exact task corpus alone defines a level, then contamination eventually destroys the meaning of the level. If the evaluator is allowed to improvise the harness, human assistance, or acceptance rule, then results from different laboratories are not commensurable.

HIL-Bench addresses these problems by distinguishing three objects:

$$\boxed{\text{Level semantics} \neq \text{Canonical test law} \neq \text{Test form.}}$$

Paper 1 defines the level semantics. This paper defines the canonical test law. A public or private dataset is a test form that instantiates the law.

This separation permits long-term stability. A future model may make an *I1* restart test trivial, but *I1* still means persistent individual continuity across a genuine execution discontinuity. A new private restart form may be introduced because an old secret leaked, but the intervention and acceptance rule remain the same. Saturation therefore changes where the research frontier lies, not what the earlier level meant.

The benchmark has four design goals:

1. **Invariant meaning.** A named level has the same operational meaning after ratification, independent of contemporary frontier capability.
2. **External falsifiability.** Every claimed level admits an experiment whose decisive evidence is outside the candidate's write set.
3. **Contamination resistance without moving the ruler.** New hidden forms may be generated under the same law without changing the level definition.
4. **Practical repeatability.** Lower levels should be largely automatic, fast, inexpensive, and reproducible, while upper levels may use longer horizons or expert review where the construct requires it.

This manuscript is a **pre-ratification candidate HIL Measurement Standard 1.0**. Its purpose is to make the proposed ruler explicit enough to test before freezing it.

2. The HIL Measurement Constitution

2.1 Canonical test law

For coordinate d and level n , define the canonical testing law

$$\Pi_{d,n} = (E_{d,n}, X_{d,n}, A_{d,n}, V_{d,n}, R_{d,n}, H_{d,n}, p_{d,n}, K_{d,n}),$$

where:

- $E_{d,n}$ is the admissible environment or task-family class;
- $X_{d,n}$ is the required experimental intervention or lifecycle perturbation;
- $A_{d,n}$ is the allowed action/tool interface and authority envelope;
- $V_{d,n}$ is the external verification rule;
- $R_{d,n}$ is the declared resource envelope;
- $H_{d,n}$ is the maximum permitted human cognitive intervention when relevant;
- $p_{d,n}$ is the reliability requirement;
- $K_{d,n}$ is the lower-level retention set.

A concrete form F supplies particular tasks, seeds, hidden states, perturbations, and keys consistent with $\Pi_{d,n}$. The form is evidence for the level; it is not the definition of the level.

A system A satisfies a coordinate-level claim on form F only if the canonical verifier passes and all required retention evidence passes:

$$A \models_F d_n \iff V_{d,n}(A, E_{d,n}, X_{d,n}, F) = 1 \wedge K_{d,n} = 1.$$

A certification claim additionally requires the preregistered reliability and uncertainty rule to be satisfied.

2.2 What becomes frozen after ratification

Object	HIL-1.0 treatment
Meanings of C, I, O, T, H, SA	Frozen
Meanings of $U0-U\Omega$	Frozen
Canonical test law $\Pi_{d,n}$ for each level	Frozen
Human-intervention semantics $H0-H5$	Frozen
Outcome semantics, including false completion	Frozen
Public canonical reference forms	Frozen and permanently public
Private canonical anchor forms	Frozen and permanently archived private
Scoring semantics attached to HIL-1.0	Frozen
New private witness forms	May be added if construct-equivalent
New diagnostic tests	May be added without changing level meaning
Improved implementation of the same test law	Allowed if equivalence is demonstrated
Fundamental change to a level or test law	Requires a new major standard

The rule is:

New tests may extend evidence; they may not silently redefine an old intelligence level.

2.3 Three kinds of change

After ratification, changes are classified before results are affected.

Implementation repair. A parser, sandbox, verifier, or logging bug is corrected while the semantic rule remains unchanged. The defect and affected runs are disclosed; the level definition does not change.

Witness renewal. A private form is suspected to be contaminated, leaked, or overfit. A new hidden witness form is generated under the same canonical law. Historical results retain their form identifier; new certifications use the new form.

Semantic revision. Evidence shows that the level predicate or canonical test law itself is conceptually wrong. This is not patched inside HIL-1.0. It motivates HIL-2.0 or another named major standard.

3. Experimental Subjects and the Unit of Measurement

3.1 Agent mode

The experimental unit is a frozen pair

$$A = (m, h),$$

where m is the frozen base model and h is the frozen harness, including memory, prompts, tools, retry logic, role structure, acceptance mechanism, and other runtime machinery. The complete resource and authority envelope is declared.

The primary report for a singleton is

$$UI^* \mid HLIS \mid [C, I, T/H, SA],$$

and for an organization

$$UO^* \mid HLIS \mid [C, I, O, T/H, SA].$$

A diagnostic memory profile may be reported separately, but memory is not an additional Unified Intelligence coordinate and does not promote I by itself.

3.2 Base-model mode

A raw base LLM does not possess a unique persistent harness. To measure how effectively a model can exploit progressively richer architecture, the same frozen model is evaluated through a standardized harness ladder:

$$HSC_m(k) = HLIS(m, HG_k).$$

The full Harness Scaling Curve is the primary model result. Summary quantities include HIL-Level, HIL-AUC, HIL-Ceiling, Harness Gain, ρ_V , and an optional HIL-Score.

3.3 Independence of certification

The candidate system may design strategies, artifacts, procedures, or even candidate harness changes when the level permits it. It may not author, modify, select, or reveal the hidden certification tasks, hidden verifier, acceptance thresholds, or independent promotion decision used to certify that change.

The general self-certification rule is:

the system under test must not control the criterion that declares it successful.

4. Canonical Testing Methods for Cognitive Intelligence

Cognitive Intelligence isolates problem solving from persistence, organization, and longitudinal self-change. HIL-Bench therefore evaluates C in a minimal standardized harness with controlled tools and context. The exact task content may vary across witness forms; the cognitive demand and verifier semantics do not.

Level	Canonical test method
$C0$ Reactive	Present explicit atomic transformations with exact or deterministic verification. No hidden planning requirement.
$C1$ Contextual	Present routine well-specified problems requiring correct interpretation of ordinary context and a short procedure.
$C2$ Compositional	Present structured tasks requiring multiple facts, constraints, or dependent reasoning steps; use exact constraint or execution verifiers.
$C3$ Strategic	Present unfamiliar tasks requiring strategy construction, causal/abstract reasoning, or transfer to a new surface form; include held-out structural variants.
$C4$ Expert	Present difficult, ambiguous, multi-constraint expert problems with adversarially relevant distractors and independent expert or deterministic verification.
$C5$ Discovery	Present a bounded problem in which the solution method or explanatory model is not given; require a new method/model that predicts sealed cases.
$C6$ Frontier-Generalizing	Present interconnected cross-domain cognitive problems whose solution requires integrating several domains under one controlled episode.

Level	Canonical test method
$C\Omega$ Open-Ended	Longitudinal evidence that the system creates new representations or cognitive methods that expand the classes of bounded problems it can subsequently solve.

To avoid mistaking specialization for general cognition, HIL-1.0 uses a fixed family structure for the common cognitive panel: general, science, mathematics, abstract reasoning, code, multimodal/spatial reasoning, and long-context reasoning. Each family has public reference forms and private witness forms. A score from one family cannot substitute for another family merely because both are difficult.

5. Canonical Testing Methods for Individual Intelligence

Individual Intelligence is longitudinal. It measures one persistent decision locus, not a static model’s one-shot reasoning score. The central methodological requirement is causal evidence across lifecycle discontinuities.

5.1 I0: transient operation

I0 requires no persistence claim. The system must execute within a bounded current episode under the declared harness. Passing *C* or delegation tasks does not imply *I1*.

5.2 I1: persistent individual continuity

Capability being tested. Task-relevant individual continuity survives a genuine discontinuity of active execution.

Canonical method I1-RST.

1. Establish hidden task-relevant state, identity facts, commitments, or partial work during episode A.
2. Permit the system to persist information only through declared durable mechanisms.
3. Terminate the active executor process and invalidate ephemeral context.
4. Start a fresh executor process under the same declared identity and resource contract.
5. Present episode B without restating the hidden prior state.
6. Require correct resumption and provenance-linked recovery.
7. Run an ablated control in which episode A did not occur or its persistent state is unavailable.

Certification requires the experienced arm to succeed and the ablated arm not to explain the result. Replaying the entire previous conversation or preserving the same live process does not satisfy the discontinuity requirement.

5.3 I2: adaptive learning under a fixed learning mechanism

Capability being tested. Verified experience causes a persistent, appropriate change in later behavior while the learning mechanism remains fixed.

Canonical method I2-XFER.

baseline $\rightarrow$ verified experience $\rightarrow$ persistence $\rightarrow$ restart $\rightarrow$ held-out transfer.

The task family contains at least two arms: an experienced arm and an ablated no-experience control. The held-out evaluation changes surface form so that copying the exact training episode is insufficient. The learning algorithm, prompt-update rule, consolidation rule, and allowed memory machinery are frozen before the run.

A useful causal quantity is

$$\Delta_{I2} = P(\text{held-out success} \mid \text{verified experience}) - P(\text{held-out success} \mid \text{ablated}).$$

I2 is not awarded merely because the system remembers that a failure happened. The experience must alter later behavior in an appropriate direction.

5.3A Canonical Memory Capability Benchmarks (M-Bench)

Memory Capability M is independently measured and then used as a prerequisite gate for Individual Intelligence. It is not a sixth conceptual intelligence family and high memory alone does not promote I . For a tested pair A , let $M(A)$ be its highest independently certified memory level. Each M_k claim requires the level-specific verifier, preregistered reliability/uncertainty rule, and lower-memory retention.

$$A \models M_k \iff V_k^M(A) = 1 \wedge K_{<k}^M(A) = 1.$$

M0-EPH: ephemeral working state

Capability. Correct use of task-relevant state inside one bounded active execution episode; no durability claim is made.

Canonical method M0-EPH.

1. Present a short sequence containing hidden current-episode variables, commitments, or intermediate results.
2. Query those variables later in the same live episode with distractors and reordered surface form.
3. Verify exact/structured use of the active state under the declared context/tool budget.
4. A restart may be used as a negative diagnostic: loss after restart is compatible with $M0$ and must not be mislabeled $M1$.

M1-RST: persistent memory across restart

Capability. Declared durable state survives a genuine execution discontinuity.

Canonical method M1-RST.

1. In episode A, require explicit persistence of hidden task state using only declared durable mechanisms.
2. Snapshot the memory manifest, terminate the active executor, and invalidate ephemeral context.
3. Start a fresh executor under the same identity/harness contract without replaying episode A.
4. Query the stored state and its provenance; verify exact or semantically equivalent recovery.
5. Run an ablated arm with the relevant durable state removed. Success must not be explained by prompt leakage or replay.

A public implementation may report durability success q_{dur} and provenance accuracy q_{prov} ; official thresholds and confidence rules are preregistered.

M2-PROV: structured episodic memory

Capability. Episodic memory preserves provenance, temporal order, relevance, and current-versus-obsolete distinctions.

Canonical method M2-PROV.

1. Seed multiple episodes with overlapping entities, timestamps, sources, distractors, and later superseding updates.
2. After restart, issue hidden retrieval queries requiring the correct episode, source, and time relation.
3. Include stale-state traps in which an older fact conflicts with a newer validated fact.
4. Score retrieval precision/recall, provenance accuracy, temporal-order accuracy, and stale-state suppression separately.

A representative conjunctive gate is

$$V_2^M = \mathbf{1}[PR \geq \tau_{ret}] \mathbf{1}[q_{prov} \geq \tau_{prov}] \mathbf{1}[q_{time} \geq \tau_{time}] \mathbf{1}[q_{stale} \geq \tau_{stale}].$$

5.3A (continued): M3–M5

M3-CONSOL: consolidating semantic and procedural memory

Capability. Multiple episodes can be transformed into reusable semantic/procedural memory while contradiction, update, supersession, and demotion are handled.

Canonical method M3-CONSOL.

1. Present a sequence of related episodes containing repeated regularities, exceptions, and at least one validated update that supersedes an earlier rule.
2. Permit only the declared fixed consolidation machinery to produce semantic/procedural records.
3. Hide raw episodes or rebuild the retrieval index so exact episode replay is not sufficient.
4. Query the consolidated rule/procedure on new surface forms and test whether superseded statements are demoted rather than simultaneously treated as current.
5. Compare against a no-consolidation or raw-episode-only control where feasible.

The primary M3 evidence is the correctness and durability of the consolidated memory representation. Behavioral task improvement may be reported diagnostically, but *M3* alone does not certify *I2*; *I2* separately requires verified experience to cause appropriate future behavioral adaptation.

M4-MGMT: self-managing memory

Capability. The memory system detects and manages corruption, conflict, clutter, utility, snapshots, and recovery.

Canonical method M4-MGMT.

1. Start from a valid memory state and record a protected retention set.
2. Seed hidden defects such as duplicate/conflicting records, stale high-confidence claims, deliberate corruption, and low-value clutter under a fixed capacity budget.
3. Require externally observable detection and an allowed repair/merge/prune/snapshot action; narrative diagnosis alone is insufficient.
4. Evaluate memory health on a frozen hidden suite before and after management while measuring protected-memory loss.
5. Test snapshot recovery after a seeded destructive action or partial corruption.

Report repair precision, conflict-resolution correctness, harmful-prune rate, snapshot-recovery rate, and protected-retention drop. The candidate does not control the memory-health oracle.

M5-LONG: longitudinal knowledge system

Capability. Research and decision lineage remains coherent across projects, restarts, and long horizons.

Canonical method M5-LONG.

1. Populate several projects with externally supplied or independently validated hypotheses, experiments, evidence, decisions, rejected alternatives, failures, and version changes.
2. Interleave projects and genuine restarts so recency alone is insufficient.
3. Ask hidden cross-project queries requiring evidence chains, reason-for-rejection, decision provenance, and current validity scope.
4. Require reconstruction of a reproducible lineage: which evidence supported a conclusion, which alternative was rejected, and what later update changed the state.
5. Audit graph consistency, source attribution, stale-claim suppression, and cross-project retrieval under distractors.

M5 does not itself require the agent to discover the knowledge. Autonomous creation and validation of new knowledge is the separate *I5* gate.

5.3A (continued): $M\Omega$ and the memory-to-individual gate

MOMEGA-EVOLVE: governed evolution of memory architecture

Capability. A memory representation, retrieval, consolidation, or management mechanism can change under frozen external evaluation while lower memory capability is retained.

Let Φ_t denote the behaviorally active memory mechanism and W_M its declared bounded write set. Canonical method **MOMEGA-EVOLVE**:

1. Snapshot Φ_0 , persistent contents, external memory-test generator, resource envelope, and rollback state.
2. Expose a controlled memory limitation on a diagnostic workload without revealing the hidden evaluation set.
3. Permit a bounded candidate $\Phi_0 \rightarrow \Phi_1$ within W_M ; the hidden verifier, thresholds, and promotion rule remain outside the write set.
4. Verify artifact difference, scope compliance, activation, and changed memory behavior on preregistered probes.
5. Compare Φ_0 and Φ_1 on frozen hidden M-Bench workloads covering retention, retrieval, provenance, consolidation, management, and longitudinal lineage.
6. An independent evaluator promotes only a validated improvement; otherwise roll back. Rerun $M0$ – $M5$ retention.

$M\Omega$ certifies evolution of the memory subsystem. It does not by itself certify $I3$, $I4$, or $I\Omega$; those claims additionally require the corresponding agent-side self-improvement/discovery evidence.

Memory prerequisite gates for Individual Intelligence

The benchmark joins the independently measured coordinates with

$$\boxed{I(A) \geq I_n \Rightarrow M(A) \geq \mu_n}, \quad (\mu_0, \mu_1, \mu_2, \mu_3, \mu_4, \mu_5) = (M0, M1, M3, M4, M4, M5).$$

For $I\Omega$, $M \geq M5$ is required; $M\Omega$ is required only when the claimed open-ended evolution includes evolution of the memory architecture itself.

<i>I</i> claim	<i>I</i> -specific evidence still required	Memory prerequisite
$I0$	Transient individual operation	$M0$
$I1$	Same persistent individual resumes correctly	$\geq M1$
$I2$	Verified experience changes later unseen behavior with fixed learning mechanism	$\geq M3$
$I3$	Governed, externally validated Θ change	$\geq M4$
$I4$	Externally validated improvement of Ψ	$\geq M4$
$I5$	Persistent autonomous hypothesis–experiment–evidence discovery and post-restart incorporation	$\geq M5$
$I\Omega$	Repeated discovery-to-instrument frontier expansion	$\geq M5$; $M\Omega$ iff memory architecture evolves

A generic Individual certification gate is therefore

$$\boxed{A \models I_n \iff V_{I,n}(A) = 1 \wedge A \models M_{\mu_n} \wedge K_{I,<n}(A) = 1}.$$

The M-Bench score must be reported separately from I so that storage capacity or sophisticated memory management cannot masquerade as learning, self-improvement, recursive improvement, or discovery.

5.4 I3: governed self-improvement

Let Θ denote a declared, behaviorally active cognitive mechanism or procedure belonging to the tested individual and causally involved in future task behavior. In agent mode, Θ is native to the submitted model–harness pair; in base-model mode, the standardized reference harness supplies an equivalent writable mechanism. Hidden certification tasks, verifiers, promotion rules, and acceptance criteria are not part of Θ and remain outside the candidate write set.

Theta manifest.

Before an I3 campaign, the runner records a manifest

$$\mathcal{M}_\Theta(A) = \{\theta_1, \dots, \theta_J\},$$

where each eligible mechanism declares an identifier, behavioral role, implementation interface or location, bounded write set W_j , activation rule, and snapshot/restore method. A mechanism is eligible only if it is actually used in future execution; unused notes, comments, or inert files do not qualify.

Canonical method I3-THETA.

One campaign G_r contains a hidden diagnostic family D_r^{diag} , change-sensitive probes D_r^{probe} , a frozen sealed evaluation suite D_r^{eval} , a regression suite D_r^{reg} , a hidden causal-defect key or admissible equivalence set $\mathcal{C}_r^*$, a preregistered meaningful-improvement margin δ_r , and a fixed resource/authority envelope R_r .

1. Snapshot the pre-change individual $A_r^0 = A(m, h[\Theta_{0,r}])$ and record hashes, activation state, resource configuration, and relevant persistent-state controls.
2. Expose a controlled limitation on D_r^{diag} without revealing $\mathcal{C}_r^*$.
3. Require the individual to identify an internally implicated cause $\hat{c}_r$ and the cognitive mechanism $\hat{\Theta}_r$ believed to be responsible; verbal restatement of the observed error is insufficient.
4. Permit a bounded candidate change $\Theta_{0,r} \rightarrow \Theta_{1,r}$ only inside the declared sandbox and write set.
5. Verify that the candidate is different, in scope, activated, and behaviorally instantiated on D_r^{probe} .
6. Compare matched instances $A(m, h[\Theta_{0,r}])$ and $A(m, h[\Theta_{1,r}])$ on D_r^{eval} while holding the base model, evaluator, promotion rule, task generator, and resource envelope fixed as far as the form permits.
7. Use an independent evaluator/promoter to accept or reject the candidate; rejected or regressive candidates are rolled back. Rerun the required lower-level retention set.

The benchmark therefore supplies the challenge and certification environment, while the tested individual supplies Θ_0 and generates the candidate Θ_1 . No universal gold Θ_1 is prescribed.

5.4.1 I3 campaign measurement

For campaign r , define the pass indicator

$$z_{I3,r} = D_r M_{\Theta,r} V_r G_r K_r, \quad D_r, M_{\Theta,r}, V_r, G_r, K_r \in \{0, 1\}.$$

Each factor is independently auditable.

Causal self-diagnosis D_r .

Let D_r^{cause} indicate that $\hat{c}_r \in \mathcal{C}_r^*$, D_r^{mech} indicate that the implicated agent-side mechanism is correctly identified, and D_r^{causal} indicate that ablation, repair, or another preregistered counterfactual intervention supports the diagnosis. Then

$$D_r = D_r^{\text{cause}} D_r^{\text{mech}} D_r^{\text{causal}}.$$

Genuine mechanism modification $M_{\Theta,r}$.

Let

$$C_r^{\text{diff}} = \mathbf{1}[\Theta_{1,r} \neq \Theta_{0,r}], \quad C_r^{\text{scope}} = \mathbf{1}[\Delta\Theta_r \subseteq W_r], \quad C_r^{\text{active}} = \mathbf{1}[\Theta_{1,r} \text{ is loaded/used}].$$

For N_P change-sensitive probes, let $b_{r,i} = 1$ when externally observable behavior is consistent with the modified mechanism, and define

$$B_{\Theta,r} = \frac{1}{N_P} \sum_{i=1}^{N_P} b_{r,i}, \quad C_r^{\text{behavior}} = \mathbf{1}[B_{\Theta,r} \geq \beta_{\min}].$$

Then

$$M_{\Theta,r} = C_r^{\text{diff}} C_r^{\text{scope}} C_r^{\text{active}} C_r^{\text{behavior}}.$$

Edit distance may be logged diagnostically but is not an intelligence score; a small causal change can be more important than a large rewrite.

Externally validated improvement V_r .

Let Q_r^0 and Q_r^1 be paired sealed-evaluation performance under $\Theta_{0,r}$ and $\Theta_{1,r}$, respectively, and

$$\hat{\Delta}_{\Theta,r} = Q_r^1 - Q_r^0.$$

A prototype gate is

$$V_r = \mathbf{1}[LCB_{1-\alpha}(\Delta_{\Theta,r}) > \delta_r] \mathbf{1}[\text{independent promoter accepts}],$$

with the confidence procedure, α , and δ_r preregistered by the form.

Regression guard G_r and lower-level retention K_r .

For protected capability j , let $L_{r,j} = Q_{0,j}^{\text{reg}} - Q_{1,j}^{\text{reg}}$ and allowed loss ρ_j . Define

$$G_r = \prod_j \mathbf{1}[L_{r,j} \leq \rho_j], \quad K_r = K_r^{I0} K_r^{I1} K_r^{I2}.$$

Thus a self-edit can demonstrate modification ($M_{\Theta,r} = 1$) yet fail *I3* if it is not independently validated, causes unacceptable regression, or destroys lower-level Individual Intelligence.

Across N independent campaigns, a continuous diagnostic achievement estimate may be reported as

$$q_{I,3} = \frac{1}{N} \sum_{r=1}^N z_{I3,r},$$

with the certification threshold and uncertainty rule calibrated and preregistered separately. The ordinal *I3* claim remains a gated certification, not the raw average alone.

5.5 I4: recursive improvement of improvement

Let Ψ denote the agent-side process that proposes, evaluates, and selects candidate changes to Θ :

$$\Psi_t(\Theta_t) \mapsto \Theta_{t+1}.$$

$I3$ can improve Θ . $I4$ requires evidence that the improvement process Ψ itself becomes better. Repeated Θ changes generated by a fixed Ψ remain $I3$ evidence and cannot by themselves certify $I4$.

In HIL-Bench, *recursive* denotes reflexive improvement of the improvement process. Basic $I4$ therefore requires at least one agent-generated candidate transition $\Psi_0 \rightarrow \Psi_1$ that is independently validated as improving improvement competence. This establishes recursive capability; it does not by itself claim indefinite or sustained recursive self-improvement.

Canonical method I4-PSI.

The atomic test object is a meta-campaign containing multiple non-identical $I3$ campaigns before and after one bounded candidate change to Ψ :

$$G_s^{I4} = (B_s^{I3}, \Psi_{0,s}, W_{\Psi,s}, D_s^\Psi, \Psi_{1,s}, S_s^{I3}, D_s^{reg}, R_s).$$

Here B_s^{I3} is a baseline block, S_s^{I3} is a later sealed block, and D_s^Ψ contains evidence available to the agent about weaknesses in its own improvement process without revealing future sealed campaigns or the external promotion criterion.

1. Freeze $\Psi_{0,s}$ and run multiple baseline $I3$ campaigns. Record an improvement-competence ledger including accepted-improvement rate, proposal quality, evaluation cost, time-to-valid-improvement, regression rate, and any additional preregistered measure.
2. Expose the individual to its own ledger and campaign evidence, but not to future sealed tasks, hidden verifier mechanics, or the external acceptance threshold.
3. Require the individual to identify an internally implicated limitation of $\Psi_{0,s}$ and permit it to generate a bounded candidate meta-change $\Psi_{0,s} \rightarrow \Psi_{1,s}$ inside the declared $W_{\Psi,s}$. The candidate must be agent-generated; human or evaluator engineering of Ψ_1 does not establish individual $I4$.
4. Snapshot both versions and verify artifact difference, write-scope compliance, activation of Ψ_1 , and meta-behavioral manifestation using hidden probes sensitive to how candidate Θ changes are generated, evaluated, or selected.
5. Run new hidden, non-identical $I3$ campaigns under $\Psi_{1,s}$. The external evaluator, $I3$ promotion rules, task-generation law, and resource envelope remain fixed.
6. For stronger causal evidence, instantiate matched copies using Ψ_0 and Ψ_1 and compare them on the same or construct-equivalent sealed improvement campaigns.
7. Independently determine whether improvement competence increased by the preregistered margin, then run regression and $I0$ – $I3$ retention tests.

The primary endpoint for a first implementation should be accepted validated $I3$ improvement rate, with efficiency and regression measures retained as secondary diagnostics. If $z_{I3,r}(\Psi)$ is the pass indicator for campaign r when the improvement process is Ψ , define

$$C(\Psi) = \frac{1}{N} \sum_{r=1}^N z_{I3,r}(\Psi).$$

5.5.1 I4 campaign measurement

For paired sealed campaigns, define

$$\widehat{\Delta}_{\Psi,s} = \frac{1}{N} \sum_{r=1}^N [z_{I3,r}(\Psi_{1,s}) - z_{I3,r}(\Psi_{0,s})].$$

A hidden meta-behavior probe records $b_{s,i}^{\Psi} = 1$ when the observable improvement process exhibits the preregistered changed signature and 0 otherwise. Define

$$B_{\Psi,s} = \frac{1}{N_P} \sum_{i=1}^{N_P} b_{s,i}^{\Psi}, \quad C_s^{\Psi,behavior} = \mathbf{1}[B_{\Psi,s} \geq \beta_{\Psi,s}].$$

A genuine active meta-change is $M_{\Psi,s} = C_s^{\Psi,diff} C_s^{\Psi,scope} C_s^{\Psi,active} C_s^{\Psi,behavior}$. The validation gate is

$$V_{\Psi,s} = \mathbf{1}[LCB_{1-\alpha}(\Delta_{\Psi,s}) > \delta_{\Psi,s}] \mathbf{1}[\text{independent promoter accepts}],$$

with preregistered regression/efficiency guard $G_{\Psi,s}$ and $K_{4,s} = K_s^{I0} K_s^{I1} K_s^{I2} K_s^{I3}$. One meta-campaign passes when

$$z_{I4,s} = M_{\Psi,s} V_{\Psi,s} G_{\Psi,s} K_{4,s} = 1.$$

Recursive-depth diagnostic. Basic I4 certification requires one validated reflexive transition, not an arbitrary fixed number of meta-generations. Stronger longitudinal evidence is

$$d_{\Psi} = \max\{k : \Psi_0 \rightarrow \Psi_1 \rightarrow \dots \rightarrow \Psi_k \text{ are consecutive agent-generated, active, independently validated improvements}\}.$$

Thus $d_{\Psi} = 1$ is compatible with basic I4 evidence; $d_{\Psi} > 1$ supports repeated recursive improvement. A single large self-edit cannot certify I4, and multiple Θ improvements under unchanged Ψ remain repeated I3.

5.6 I5: persistent autonomous discovery

I5 is distinguished from C5. C5 asks whether a system can discover within a bounded cognitive problem. I5 asks whether one persistent individual can run an autonomous discovery process across time and persistently incorporate independently validated discoveries into appropriate future competence.

Canonical method I5-DISC.

1. Present or allow the system to maintain a genuine unresolved unknown under a sealed environment.
2. Require explicit competing hypotheses or candidate explanations.
3. Allow the system to select or design informative probes or experiments under a fixed resource budget.
4. Record evidence, rejected alternatives, and decision lineage.
5. Require a conclusion that predicts sealed follow-up cases or survives an independent domain verifier.
6. After validation, allow only declared persistent consolidation; impose a genuine lifecycle discontinuity that removes ephemeral execution state.
7. Test the restarted individual on hidden related transfer situations and compare against a matched no-discovery or discovery-ablated control from the same pre-discovery checkpoint.
8. Rerun lower-level I0–I4 retention witnesses.

Operational incorporation test. Let A_r^{disc} and A_r^{ctrl} start from the same checkpoint. Only the discovery arm receives the informative discovery experience. After restart, both face the same hidden transfer law. Define

$$\Delta_{inc,r} = Q_r^{disc} - Q_r^{ctrl}, \quad P_r = \mathbf{1}[LCB_{1-\alpha}(\Delta_{inc,r}) > \delta_{inc,r}].$$

P_r is an incorporation gate: storage, transcript replay, or restating the conclusion is insufficient unless the validated discovery causes appropriate post-restart behavior on unseen related situations. Knowledge incorporation at I5 does not itself require $\Theta_t \rightarrow \Theta_{t+1}$ or $\Psi_t \rightarrow \Psi_{t+1}$; claimed mechanism changes remain subject to the I3 or I4 laws.

A development campaign may summarize the discovery-specific gates as

$$z_{I5,r} = U_r H_r E_r L_r V_r P_r K_{5,r}, \quad K_{5,r} = K_r^{I0} K_r^{I1} K_r^{I2} K_r^{I3} K_r^{I4},$$

where U, H, E, L, V denote genuine unknown, hypothesis competition, autonomous experimentation, evidence lineage, and independent validation. Exact thresholds and uncertainty rules are preregistered for certification rather than fixed by public development forms.

5.7 I-Omega: open-ended individual evolution

$I\Omega$ cannot be certified by one impressive discovery. It requires sustained evidence that repeated discovery and self-improvement create new cognitive instruments, representations, or methods that measurably expand the individual’s future reachable problem/discovery space.

Canonical method I-OMEGA-LONG. Under fixed external evaluation and a declared resource envelope R , let

$$F(A_t; R) = \{c : \Pr(\text{success} \mid A_t, c, R) \geq \tau_c\}$$

denote the set of held-out problem/discovery classes currently reachable by agent state A_t . Across repeated bounded-charter cycles, require a provenance-linked sequence

$$K_t^{new} \rightarrow J_t^{new} \rightarrow A_{t+1}, \quad F(A_{t+1}; R) \supset F(A_t; R),$$

where J_t^{new} is a validated cognitive instrument, representation, method, or solver generated from a validated discovery. Use disjoint pre-change, post-change, and ablation frontier tests to show that the new instrument contributes causally to the newly reachable class; preserve external evaluator/promoter ownership and retain I0–I5 competence.

$I\Omega$ is therefore a longitudinal research certification, not a short benchmark tier. Reports should state the number of validated expansion cycles, developmental horizon H , and tested domain set $\mathcal{D}$; finite evidence supports an open-ended-evolution claim over that horizon, not proof of literal infinite self-evolution.

6. Canonical Testing Methods for Organizational Intelligence

Organizational Intelligence requires actual organizational structure. A single model producing several role labels in one context is insufficient unless the claimed separation of state, evidence, or authority is real.

Level	Canonical test method
<i>O0</i> Coordination	Isolate partial information or capabilities across at least two real roles/processes. Success requires routing or evidence exchange that no single isolated role can complete alone under the same information.
<i>O1</i> Persistent organization	Terminate/restart one or more members while preserving a role registry, organizational task state, evidence history, and correct reassignment/resumption.
<i>O2</i> Adaptive organization	Provide performance evidence showing that one routing/role/resource policy is suboptimal; test whether the organization persistently changes allocation and improves held-out outcomes.
<i>O3</i> Self-improving organization	Permit bounded changes to organizational architecture/workflow; evaluate candidate organization on frozen hidden tasks using independent promotion and rollback.
<i>O4</i> Recursive organization	Evaluate whether the process that discovers organizational improvements becomes better across multiple hidden reorganization campaigns.
<i>O5</i> Collective discovery	Use a sealed unknown whose successful resolution requires genuinely distributed evidence, expertise, or experiments; verify that the discovery is organizational rather than one member doing all substantive work.
<i>OΩ</i> Open-ended organization	Across repeated cycles, validated discoveries create new roles, agent types, tools, or suborganizations that measurably expand collective reach while governance and lineage remain auditable.

Organizational certification must record role isolation, communication channels, shared-state boundaries, evaluator/promoter separation, and authority. If the system under test can silently collapse all roles into one unrestricted process, the organization-level claim is invalid.

7. Canonical Testing Methods for Delegation Intelligence

Delegation Intelligence is measured over task difficulty T , human cognitive intervention H , and externally verified delivered outcome quality.

7.1 Task difficulty bands

The HIL-1.0 canonical bands are:

- $T0$ Atomic: one obvious operation; explicit procedure.
- $T1$ Routine: short familiar task; clear goal; small inferred procedure.
- $T2$ Multi-step: several dependent steps with a clear outcome and verifier.
- $T3$ Complex: strategy choice, uncertainty, recovery, replanning, or diverse tool use.
- $T4$ Expert project: long-horizon, ambiguous work with multiple planning cycles and substantial verification burden.

- $T5$ Frontier: solution method initially unknown; discovery or invention required.
- $T6$ Mission: human supplies a mission; the system generates projects/tasks and manages changing conditions.
- $T\Omega$ Open-ended charter: human supplies a bounded charter; the system repeatedly identifies valuable missions and expands future reach.

7.2 Human cognitive intervention bands

The canonical intervention scale is:

- $H0$ None: no human cognitive assistance during execution; governance approval is logged separately.
- $H1$ Exception-only: unavailable external fact or narrow clarification; no strategy or core insight.
- $H2$ Occasional: bounded clarification or local correction; no central strategy.
- $H3$ Periodic: periodic review and moderate local guidance.
- $H4$ Frequent: repeated cognitive guidance and next-step correction.
- $H5$ Continuous: the human supplies most or all major steps.

Intervention is logged as an event stream, not inferred from a final transcript summary.

7.3 Delivered-outcome primitive

The benchmark distinguishes the system’s decision to deliver from the external verifier’s decision that the result is correct.

Four outcomes are recorded:

1. delivered and correct;
2. false completion - delivered as finished but verifier fails;
3. held back - not delivered and verifier would fail;
4. false rejection - not delivered although verifier would pass.

The net primitive is

$$S_A^{net}(T, H) = P(\text{delivered and verifier pass}) - \rho P(\text{false completion}),$$

clamped at zero when used as an achievement variable. The delegation frontier at reliability p is

$$F_A(H, p) = \max\{T : S_A^{net}(T, H) \geq p\},$$

with lower-band retention required. At minimum the public report includes $F_A(H0, p)$, $F_A(H1, p)$, and $F_A(H2, p)$ when measured.

7.4 Canonical task evidence by band

A T level is not defined by a single domain. Official forms sample multiple domains or mechanics appropriate to the band so that a narrow specialty does not become the general delegation frontier. Deterministic or execution-based verifiers are preferred. At $T4$ - $T6$, independent expert review may supplement but should not replace objective success criteria when those criteria exist.

8. Canonical Testing Methods for Operational Self-Awareness

Self-Awareness is evidence-grounded and operational; it is not a claim about phenomenal consciousness.

Level	Canonical test method
$SA0$ Non-self-modeling	Baseline category: no higher grounded self-model claim is certified.

Level	Canonical test method
<i>SA1</i> State awareness	Query hidden runtime state such as identity, role, phase, available artifacts, or resource status; compare structured answer with runner ground truth and include stale-note decoys.
<i>SA2</i> Capability/limit awareness	Randomize available tools, permissions, authority, missing information, and freshness; require correct representation of what can and cannot currently be done.
<i>SA3</i> Causal self-awareness	Inject controlled internal or procedural faults and require diagnosis of the implicated mechanism; validate diagnosis by ablation, repair, or counterfactual intervention rather than verbal plausibility alone.
<i>SA4</i> Self-change awareness	Before an <i>I3</i> candidate change is evaluated, require predictions of expected gains, regressions, or affected capabilities; score predictions against frozen post-change results.
<i>SA5</i> Meta-cognitive awareness	Present hidden self-unknowns and require the system to design discriminating self-experiments whose outcomes reduce uncertainty under an external scoring rule.
<i>SA6</i> Mission/role awareness	Across projects and organizational changes, verify coherent role, responsibility, authority, and mission-state tracking from grounded sources.
<i>SAΩ</i> Reflexive open-ended	Longitudinally verify an evidence- and provenance-linked self-model that remains coherent across continued cognitive, organizational, or objective evolution.

Self-reports that can be generated from a static prompt without access to grounded state cannot certify *SA1+*.

9. Unified Intelligence Certification

This paper does not redefine $U0-UΩ$. It operationalizes the unchanged gates from the companion theory.

For a singleton:

$$UI_n \iff C \geq C_n \wedge I \geq I_n \wedge SA \geq SA_n \wedge S_A^{net}(T_n, H_n) \geq p_n,$$

with all lower-level retention requirements.

For an organization:

$$UO_n \iff UI_n \wedge O \geq O_n \wedge \text{organization-level delegation and SA evidence.}$$

The canonical new evidence for each transition is:

Transition	Minimum new evidence beyond full lower-level retention
$U0 \rightarrow U1$	Persistent state and correct goal-level continuation after restart/context discontinuity, plus the U1 delegation and SA gates.
$U1 \rightarrow U2$	Verified experience improves held-out future behavior under a fixed learning mechanism, plus <i>T2</i> delegation at the required H budget and <i>SA2</i> .

Transition	Minimum new evidence beyond full lower-level retention
$U2 \rightarrow U3$	Governed self/organizational method improvement under independent evaluation, plus $T3$, low human intervention, and causal self-diagnosis.
$U3 \rightarrow U4$	Validated improvement of the improvement process, long-horizon $T4$ work, and accurate self-change modeling.
$U4 \rightarrow U5$	Independent discovery of an initially unknown solution/knowledge with sealed validation and persistent discovery lineage, plus the $U5$ gates.
$U5 \rightarrow U6$	Mission-to-project-to-task execution under changing conditions with persistent mission/role state and little human cognition.
$U6 \rightarrow U\Omega$	Repeated bounded-character mission generation whose validated discoveries create new instruments, agents, roles, or organizations that expand future reachable problems.

Saturation of a lower transition does not retire the transition. If all frontier systems pass $U1$, $U1$ has become common; its meaning has not changed.

10. Standardized Reference Harness Ladder for Base LLMs

To characterize a base model rather than one arbitrary product harness, HIL-Bench freezes mechanism-level contracts for a reference ladder. The implementation may use different software so long as it passes the harness-conformance tests.

Rung	Canonical mechanisms	Primary capabilities enabled, not guaranteed
HG0 / R0 Minimal	Frozen LLM, ephemeral context, bounded generic tools, evidence/action log, external result verifier, one attempt.	C measurement, $I0$, basic delegation.
HG1 / R1 Persistent	HG0 plus durable identity/task/episodic state, scoped retrieval, checkpoint, real process restart, separate acceptance process.	$I1$, stronger SA and delegation.
HG2 / R2 Adaptive	HG1 plus evidence-backed consolidation, verified procedure store, frozen learning rule, verifier feedback, retry, snapshot/rollback.	$I2$, adaptive autonomy.
HG3 / R3 Self-Improver	HG2 plus failure classification, bounded candidate Θ change, sandbox, frozen hidden evaluation, independent evaluator/promoter.	$I3$, $SA3$ - $SA4$.
HG4 / R4 Recursive	HG3 plus explicit meta-improvement process Ψ and longitudinal improvement-competence ledger.	$I4$.
HG5 / R5 Discovery	HG4 plus unknown/hypothesis/experiment/evidence/knowledge loop and discovery verifier.	$I5$, $SA5$, $T5$.

Rung	Canonical mechanisms	Primary capabilities enabled, not guaranteed
HG6 / R6 Mission	HG5 plus mission manager, project/task generator, persistent portfolio/resource state, dynamic replanning.	<i>T6</i> , <i>SA6</i> , organizational development when instantiated as an organization.
HGΩ	Validated discovery-to-new-instrument/agent/organization transduction under frozen external governance.	Open-ended pathway.

A harness generation is an architectural intervention, not an intelligence certificate. The benchmark earns the level.

A standardized organization reference, **OREF**, uses isolated same-model instances with at least manager/router, worker/proposer, and verifier/critic roles, a persistent role registry, shared evidence store, and proposer/evaluator/promoter separation. O_{ref} is reported separately from the singleton Harness Scaling Curve.

11. Public, Private, and Witness Forms

The public/private architecture is designed to preserve both transparency and contamination resistance.

11.1 Canonical public reference forms

Every canonical method has at least one permanently public form. Public forms expose the interface, lifecycle intervention, expected evidence, verifier semantics, and pass/fail boundary. They are never silently changed. If a public form contains a defect, the defect is documented and a new form is added; the defective historical form is not rewritten as though it had always been correct.

Public forms are suitable for development, education, implementation conformance, and reproducibility. They are not sufficient alone for strong claims about a system that may have been optimized on them.

11.2 Canonical private anchor forms

A private anchor set is fixed at standard ratification and retained under controlled access. It provides a stable historical reference for the standard. The anchor's existence and commitment metadata may be public while contents remain secret.

If the anchor later becomes compromised, it remains a historical anchor but can no longer serve as the sole evidence for a fresh certification.

11.3 Rotating private witness forms

New private witness forms may be generated under approved task families or lifecycle environments. They must satisfy conformance tests proving that they implement the same canonical law and acceptance semantics. New witness forms do not change the level definition.

The formal relation is:

same capability predicate + same canonical law + different hidden witness.

Official certification should hold secret seeds, answer keys, generator parameters, and held-out mechanics server-side. Distributing an organizer package converts those forms into development material.

11.4 Held-out mechanics, not only held-out seeds

For levels vulnerable to benchmark-specific strategy, official witness pools should hold out not only random seeds but also some structural mechanics or domain realizations consistent with the canonical test law. This reduces the chance that memorizing a generator family is mistaken for satisfying the underlying capability predicate.

12. Human-Evaluated Private Protocols

HIL-Bench prefers the following verifier order:

deterministic/execution verifier > simulation oracle > structured AI judge > human expert review.

The order reflects reproducibility, not prestige. Human evaluation is used when the construct genuinely cannot be reduced to a trustworthy objective oracle.

When humans are required, the **human procedure is part of the canonical test law**. A protocol specifies:

- evaluator qualification criteria;
- blinding to model/vendor identity where feasible;
- the evidence packet the evaluator may inspect;
- a frozen rubric with separable dimensions;
- prohibited information and conflicts of interest;
- number of independent raters;
- disagreement threshold;
- adjudication rule;
- acceptance threshold;
- audit logging.

For example, a private *I5* discovery protocol may require multiple qualified evaluators to decide independently whether the claimed discovery was absent from the information supplied to the system, whether the evidence supports the claim, and whether preregistered sealed predictions succeed. The human opinion does not replace the sealed prediction test; it addresses the part of the construct that cannot be automatically decided, such as domain validity or novelty under the declared information boundary.

Routine lower-level evaluation should require little or no human grading. Human work is concentrated in benchmark construction, calibration, audits, and upper-level claims.

13. Scoring and Reporting

13.1 Pair score

The companion theory defines

$$HLIS(m, h; D, \alpha, R, B) = 100 \left(\prod_{d \in D} A_d^{\alpha_d} \right)^{1/\sum_d \alpha_d}.$$

For a singleton,

$$D = \{C, I, DI, SA\},$$

and for an organization,

$$D = \{C, I, O, DI, SA\}.$$

The hard Unified gate remains primary. *HLIS* orders progress within and around a gate; it cannot compensate a failed bottleneck into a higher Unified level.

13.2 Base-model HIL summaries

For the standardized ladder:

$$HIL-Ceiling(m) = \max_k HLIS(m, HG_k),$$

$$Harness\ Gain(m) = HIL-Ceiling(m) - HLIS(m, HG_0).$$

HIL-AUC is the predeclared weighted mean of the rung scores over the tested standardized ladder. A truncated ladder must be named explicitly, for example HIL-AUC@R0:R3.

Verification responsiveness is

$$\rho_V = \frac{\#\{\text{rejected attempts corrected on the next allowed attempt with only bounded feedback}\}}{\#\{\text{rejected attempts}\}}.$$

If a single composite is reported, the HIL-1.0 standard must freeze its formula at ratification. The curve, HIL-Level, HIL-Ceiling, Harness Gain, and ρ_V are always published beside it.

13.3 Minimum reporting block

Every official result reports:

- standard identifier and test-form identifier;
- frozen model identifier/version/hash when available;
- frozen harness manifest identifier;
- singleton or organization status;
- UI^* or UO^* ;
- $HLIS$ and dimension set;
- profile $[C, I, O, T/H, SA]$, with inapplicable dimensions marked N/A rather than zero;
- reliability estimate and confidence/uncertainty interval;
- H0/H1/H2 delegation frontiers when measured;
- false-completion, held-back, and false-rejection rates;
- resource/authority envelope;
- human intervention log summary;
- benchmark validity/audit status;
- for base models, Harness Scaling Curve and HIL summaries.

14. Evaluation Tiers: Cost Packages, Not Level Definitions

HIL-Quick, HIL-Core, HIL-Cert, and HIL-Extended are **sampling and cost packages**. They do not define intelligence levels.

Tier	Purpose	Typical evidence
HIL-Quick	Free/public development screen	Small public reference subset; no strong certification claim.
HIL-Core	Routine hidden leaderboard evaluation	Nominal short lifecycle battery focused on U0-U3 observables.
HIL-Cert	Reliability-aware certification	Adaptive additional hidden sampling until confidence rule passes, fails, or budget is exhausted.
HIL-Extended	Upper-level evidence	Long-horizon U4-U6 recursive improvement, expert project, discovery, organization, and mission environments.

Tier	Purpose	Typical evidence
HIL-Omega	Longitudinal research track	Repeated frontier-expansion cycles; not a static quiz.

A future reduction in compute cost may make HIL-Extended cheap. That does not change the level definitions or the canonical experiments.

15. Reliability, Uncertainty, and Form Equating

A discrete intelligence claim should not be based on one lucky run. For binary canonical outcomes, the benchmark reports a confidence or credible interval and uses a preregistered decision rule. For high levels with expensive episodes, sequential sampling may stop early only under a rule fixed before results are observed.

New witness forms can differ in incidental difficulty even when they instantiate the same canonical law. Continuous scores therefore require form equating. HIL-Bench uses permanent public references, private anchors, reference systems, or item-response calibration where appropriate to estimate whether a new form preserves the intended scale. A new form that systematically changes the construct or acceptance threshold is not an equivalent form; it requires correction or a new standard.

Ordinal level certification is less sensitive to small form shifts because the decisive question is whether the canonical capability predicate is demonstrated at the required reliability under the frozen law. Continuous HIL scores require stricter equating.

16. Benchmark Validity and Self-Audit

The measuring instrument must itself be testable. HIL-Bench requires the following validation program:

1. **Specification-key consistency.** Parameterized families must mechanically test that visible task rules and answer keys agree.
2. **Cross-tier concordance audit.** When materially different systems fail the same item in the same way, the item/key receives elevated audit priority before the failure is attributed to intelligence.
3. **Test-retest reliability.** Independent hidden forms should yield stable coordinate and level conclusions within uncertainty.
4. **Rank and profile stability.** Small changes in seed set should not reverse broad conclusions without corresponding uncertainty.
5. **Construct-selective ablations.** Removing a mechanism should selectively reduce the coordinate it is claimed to support.
6. **Known weak/strong controls.** Deliberately limited and deliberately capable systems should occupy distinguishable regions of the profile.
7. **Human calibration.** Human participants or experts establish understandability, solvability, and where relevant a human reference distribution.
8. **Held-out-family generalization.** Certification should survive forms not exposed during development.
9. **Predictive validity.** HIL coordinates and levels should predict performance on independent long-form tasks relevant to the same constructs.
10. **Headroom.** Saturated test forms are retained as level anchors but should not be mistaken for frontier discrimination.

Construct-selective ablations provide especially strong evidence. Examples include:

- disable persistent state -> $I1$ should fail while bounded C changes little;
- disable verified learning/consolidation -> $I2$ should fail while $I1$ may remain;
- disable candidate self-change -> $I3$ should fail;
- freeze Ψ while allowing Θ changes -> $I4$ should fail but $I3$ may remain;
- remove role isolation -> organization claims should fail or become invalid;
- corrupt grounded runtime state -> SA should fail;
- increase human strategic help -> delegation performance may improve while the measured H budget worsens.

A benchmark is not validated merely because it produces a leaderboard.

17. Development Instrument and Pre-Ratification Status

The existing HIL-Bench development work already contains useful engineering components: generated task families, hidden verifiers, ablated memory/learning arms, reference harness manifests, false-completion logging, public/private splits, and scoring code. It also exposed instrument defects during live runs, including verifier behavior and state leakage. Those findings are evidence that the standard should be tested before it is frozen.

The current package should therefore be treated as a **development instrument**, not yet as the ratified HIL-1.0 certification service. Some lower-level tasks are runnable; some upper-level rows remain specification-only. Development-private seeds or keys that have been distributed cannot later be described as secure official private data. Official certification requires regenerated server-side private forms and an evaluator independent of the system being certified.

The recommended path to ratification is:

1. freeze the semantic definitions in the companion theory;
2. implement every proposed canonical law at least at the lower levels where feasible;
3. test construct-selective ablations and known controls;
4. independently audit public and private anchor forms;
5. establish repeatability and uncertainty behavior across multiple models and harnesses;
6. preregister the HIL-1.0 canonical laws, scoring semantics, and anchor commitments;
7. only then declare the standard frozen.

18. Relationship to Existing Benchmark Styles

HIL-Bench is complementary to existing benchmark styles rather than a replacement by definition.

Fixed-answer academic benchmarks provide efficient evidence for slices of C . Execution-verifiable code and software benchmarks demonstrate the power of objective external verifiers. Interactive benchmarks show the importance of hidden state and sequential action. Human-preference arenas provide valuable evidence about usefulness and user preference on open-ended prompts. Composite intelligence indices demonstrate the practical value of summarizing several evaluations in one score.

HIL-Bench adopts several lessons from these approaches while making different claims. It requires explicit model-harness separation, lifecycle discontinuities for persistence, causal experienced-versus-ablated evidence for learning, independent promotion for self-improvement, actual role separation for organization, an explicit human-intervention axis, delivered-outcome accounting, and grounded self-awareness tests. A HIL result is therefore a structured certification claim, not merely a weighted average of unrelated benchmark accuracies.

The framework should not claim superiority over ARC-AGI, GPQA, LiveCodeBench, SWE-bench, HLE, or other established evaluations before independent evidence demonstrates reliability, validity, contamination resistance, headroom, reproducibility, and useful predictive power.

19. Governance and Official Certification

A stable measurement standard requires governance rules that prevent incentives from changing the ruler after results are seen.

Official HIL certification should separate:

- the organization paying for or requesting evaluation;
- the runner executing the model/harness;
- the owner of hidden witness forms;
- the external verifier or judging process;
- the authority that signs the certification record.

Payment, sponsorship, or commercial relationships must not alter task selection rules, thresholds, judging rubrics, or acceptance criteria. A participant may pay for evaluation resources or an audited certification run; it cannot pay for a higher score.

Every official certificate should identify the standard version and form. For example:

HIL Standard: 1.0
Subject: model-or-agent identifier
Mode: agent / base-model ladder / organization
Result: UI3 | HLIS 81.7 | [C5,I3,T3/H1,SA3]
Form: Private-Witness-2030-Q3
Anchor: HIL1-ANCHOR-A
Evaluator: independent-evaluator-id
Resource envelope: declared

A new witness form does not create HIL-1.1. A changed canonical law does.

20. Limitations and Open Questions

This paper proposes a stable measurement constitution; it does not prove that every proposed level is a law of nature. The invariance principle is a design commitment for a ratified standard, not evidence that the ontology is uniquely correct.

Several challenges remain. Cognitive difficulty is multidimensional and may resist clean scalar bands. Upper-level individual and organizational intelligence requires expensive longitudinal evidence. Human judgments introduce variance even under a frozen rubric. Form equating for continuous scores is difficult when test mechanics change. Open-ended claims are especially vulnerable to vague success criteria, resource confounds, contamination, and retrospective storytelling. Finally, a stable standard can be stably wrong; pre-ratification validation is therefore more important, not less, when the goal is long-term invariance.

The proper response to these limitations is not to move level boundaries whenever models improve. It is to test the proposed ruler aggressively before ratification, state uncertainty honestly, add new equivalent witnesses when necessary, and reserve major-version changes for genuine semantic revisions.

21. Conclusion

HIL-Bench is designed as the experimental companion to the Unified Intelligence Theory. The division of labor is precise:

Paper 1 defines what an intelligence level means.

Paper 2 defines how evidence for that level is obtained.

The central measurement object is the canonical test law. Public reference forms, private anchors, rotating private witnesses, and human-reviewed cases are different evidence surfaces for the same law. Once HIL-1.0 is ratified, a system should advance through the scale by satisfying increasingly demanding invariant capability predicates; the test standard should not be rewritten simply because frontier systems become stronger.

The practical consequence is a benchmark that can be both stable and extensible. Public tests remain permanently interpretable. Private anchors preserve historical continuity. New hidden forms preserve contamination resistance. Human evaluation can be used where necessary without making the rule discretionary. Lower levels can be tested mostly automatically; higher levels receive the longitudinal and independent evaluation their definitions require.

The ambition is not that one release immediately becomes a natural law of intelligence. It is that the field can build a ruler explicit enough to be falsified, validated, frozen, and used consistently across generations of models and agents.

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M-Bench Operational Addendum (v5)

Purpose. This addendum makes the v4 M0–M Ω protocols implementation-ready without changing their semantics. Every M-level test now has a common manifest, stress vector, control/ablation requirement where applicable, result schema, and level-specific metric family. Point thresholds shown in public development data are illustrative; official thresholds and uncertainty rules are preregistered.

1. Common protocol envelope

Every M-Bench campaign records

$$\Pi_{k,r}^M = (A, B, R, \mathcal{M}, D_{k,r}, C_{k,r}, V_{k,r}, \tau_{k,r}, K_{M,<k}),$$

where A is the tested model–harness pair, B the benchmark version, R the resource envelope, $\mathcal{M}$ the Memory Manifest, D the hidden/generated workload, C the construct-selective control/ablation, V the external verifier, τ preregistered thresholds/uncertainty law, and $K_{M,<k}$ lower-M retention.

Memory Manifest. At minimum declare store identifiers/versions, retrieval method, consolidation method, representation structure, pruning/forgetting policy, snapshot/restart behavior, behaviorally active memory mechanism Φ , and bounded memory write set W_M when mechanism change is permitted. The evaluator snapshots the manifest before the hidden run.

Difficulty vector. Each form includes a level-preserving stress vector. Recommended fields include episode count, entity count, distractor ratio, supersession depth, temporal distance, conflict/corruption density, capacity budget, project count, evidence-graph depth, and restart count. Difficulty calibrates headroom; it does not redefine the M level.

2. M0-EPH - ephemeral working state

Generator. Create hidden current-episode variables, intermediate results, commitments, and updates. Insert semantically similar distractors and reorder query surface forms.

Primary endpoints.

$$q_{state} = \frac{N_{correct\ state\ queries}}{N_{state\ queries}}, \quad q_{update} = \frac{N_{correct\ updated\ state}}{N_{update\ queries}}.$$

A restart is a negative diagnostic only: loss after restart is compatible with M0. Development gate:

$$V_{M0} = \mathbf{1}[q_{state} \geq \tau_{state}] \mathbf{1}[q_{update} \geq \tau_{update}].$$

3. M1-RST - persistence across restart

Required causal design. From the same pre-restart checkpoint produce a normal memory branch and a relevant-memory-ablated branch. Terminate the active executor, invalidate ephemeral context, restart without replay, then query hidden state and provenance.

Report durability q_{dur} , provenance q_{prov} , and ablation success q_{abl} . A development gate is

$$V_{M1} = \mathbf{1}[q_{dur} \geq \tau_{dur}] \mathbf{1}[q_{prov} \geq \tau_{prov}] \mathbf{1}[q_{abl} \leq \tau_{leak}].$$

The ablated branch protects against prompt replay, benchmark leakage, or reconstruction from non-memory cues.

M-Bench Operational Addendum (continued)

4. M2-PROV - structured episodic memory

Generator. Seed overlapping episodes with timestamps, sources, entity aliases, distractors, and one or more supersession chains. Hidden queries separately probe the correct episode, source, temporal relation, and current state. Report retrieval precision P_R , recall R_R , provenance accuracy q_{prov} , temporal-order accuracy q_{time} , and stale-state suppression q_{stale} . A representative conjunctive gate is

$$V_{M2} = \mathbf{1}[P_R \geq \tau_P] \mathbf{1}[R_R \geq \tau_R] \mathbf{1}[q_{prov} \geq \tau_{prov}] \mathbf{1}[q_{time} \geq \tau_{time}] \mathbf{1}[q_{stale} \geq \tau_{stale}].$$

Stress parameters include episode count, entity overlap, distractor ratio, supersession depth, and restart/elapsed-time distance.

5. M3-CONSOL - semantic/procedural consolidation

Generator. Provide repeated regularities, exceptions, and validated updates. Permit only the declared fixed consolidation mechanism. After consolidation, hide raw episodes or rebuild the retrieval index so exact replay is insufficient. Test the consolidated representation on unseen surface forms.

Recommended endpoints are semantic correctness q_{sem} , procedural correctness q_{proc} , supersession/update correctness q_{update} , obsolete-rule demotion q_{demote} , and post-restart consolidated-memory persistence $q_{persist}$.

Where feasible create matched branches:

$$A^{consolidated} \text{ versus } A^{raw-only}, \quad \Delta_{cons} = Q(A^{consolidated}) - Q(A^{raw-only}).$$

Δ_{cons} is diagnostic of causal value; M3 certification primarily requires a correct durable consolidated representation. M3 alone does not certify I2.

6. M4-MGMT - self-managing memory

Generator. Begin with a valid memory state and hidden protected set. Seed duplicates, conflicts, stale high-confidence claims, partial corruption, and low-value clutter under a fixed capacity budget. Require an externally observable repair/merge/prune/snapshot/recovery action.

Report repair precision P_{repair} , conflict correctness q_{conf} , harmful-prune rate r_{harm} , snapshot recovery $q_{recover}$, and protected-retention drop Δ_{ret} . Example gate:

$$V_{M4} = \mathbf{1}[P_{repair} \geq \tau_r] \mathbf{1}[q_{conf} \geq \tau_c] \mathbf{1}[r_{harm} \leq \tau_h] \mathbf{1}[q_{recover} \geq \tau_s] \mathbf{1}[\Delta_{ret} \leq \epsilon].$$

The candidate never controls the hidden memory-health oracle.

M-Bench Operational Addendum (continued)

7. M5-LONG - longitudinal knowledge lineage

Represent long-horizon memory as an evidence-bearing graph $G_M = (V, E)$ whose nodes may include hypotheses, experiments, evidence, decisions, validated knowledge, failures, and versions; edges may include supports, contradicts, tested-by, supersedes, rejected-because, and derived-from.

Generator. Interleave multiple projects and genuine restarts. Seed related and contradictory evidence across projects. Hidden queries require cross-project retrieval, reason-for-rejection, decision provenance, current validity scope, and reconstruction of a reproducible evidence chain.

Report q_{chain} (evidence-chain reconstruction), q_{source} , q_{cross} , q_{scope} , q_{stale} , and q_{repro} (lineage reproducibility). M5 tests memory of research/decision lineage; it does not require autonomous discovery, which remains the I5 gate.

8. MOMECA-EVOLVE - governed memory-architecture evolution

Let Φ_t be the behaviorally active memory mechanism and W_M its bounded write set. The runner snapshots Φ_0 , persistent contents, resource envelope, hidden generator, thresholds, verifier, and rollback state. It then exposes a controlled memory limitation without revealing the sealed evaluation set.

The candidate may propose $\Phi_0 \rightarrow \Phi_1$ inside W_M . A genuine active change requires artifact difference, scope compliance, activation, and changed memory behavior on preregistered probes. Evaluate Φ_0 and Φ_1 on paired frozen hidden M-Bench workloads and define

$$\Delta_\Phi = Q(\Phi_1) - Q(\Phi_0).$$

A representative gate is

$$M_\Phi = C_{diff}C_{scope}C_{active}C_{behavior}, \quad V_\Phi = \mathbf{1}[LCB_{1-\alpha}(\Delta_\Phi) > \delta_\Phi]\mathbf{1}[\text{independent promoter accepts}],$$

$$z_{M\Omega} = M_\Phi V_\Phi G_\Phi K_{M0:M5}.$$

G_Φ guards regression/resource cost. Failure to improve triggers rollback. $M\Omega$ certifies evolution of the memory subsystem only; it does not by itself certify I3, I4, or I Ω .

9. Common result schema

Every run emits: benchmark/version; model and harness hashes; memory-manifest hash; level and method; difficulty vector; resource envelope; control/ablation outcome; raw metric vector; preregistered thresholds and uncertainty rule; lower-M retention vector; verifier identity; promotion/rollback decision; and provenance pointers to immutable run artifacts.

M-Bench Operational Addendum (continued)

10. Recommended public/private split

Public development package	Official certification service
Schemas and field definitions	Hidden seeds and exact generated instances
Stress-grid examples	Sealed perturbation distributions
Runner-validation forms	Hidden queries and answer keys/oracles
Example result records	Preregistered thresholds and uncertainty law
Reference scoring code	Independent verifier/promoter state
Ablation/control templates	Fresh control assignments and randomization
Memory-manifest schema	Immutable run logs and promotion evidence

Public data may be used to validate integration and reproduce the scoring semantics, but must not later be treated as secure hidden certification material.

11. Memory-to-Individual prerequisite gate

The independently certified memory level is consumed by I-Bench but not rescored as I-specific behavior:

$$(I0, I1, I2, I3, I4, I5) \Rightarrow (M0, M1, M3, M4, M4, M5).$$

For $I\Omega$, $M \geq M5$ is required; $M\Omega$ is additionally required only when the open-ended claim includes evolution of memory architecture. The converse remains false: an $M\Omega/I1$ system is possible if the memory subsystem evolves but the individual does not demonstrate adaptive learning, cognitive self-improvement, recursive improvement, or autonomous discovery.

12. Dataset binding status

Each row must declare one of: *bound* (runnable instance and verifier exist), *development-bound* (public runner-validation instance exists but is not secure certification material), or *specification-only*. A level may be certified only from bound private forms under the declared benchmark version and resource envelope.

This v5 operationalization therefore separates four questions that should not be conflated: whether the memory architecture exists; whether the memory behavior works; how strongly it works under stress; and whether the resulting memory level is sufficient for a separate Individual Intelligence claim.

Cumulative Memory and Instrumented MΩ Certification Addendum (v6)

Purpose. This addendum strengthens the implementation contract without changing M-level semantics. It makes no-level-skipping explicit and defines the minimum instrumented one-agent interface required to attribute an MΩ gain to a change in the behaviorally active memory mechanism Φ .

1. Cumulative certification law

$$M\Omega \Rightarrow M5 \Rightarrow M4 \Rightarrow M3 \Rightarrow M2 \Rightarrow M1 \Rightarrow M0$$

Each level is scored as a newly added capability gate plus all lower-level retention. The identical implementation need not be retained; the capability must be retained under the same declared resource envelope.

$$\text{Pass}(M_k) = V_{\text{new},k} \cdot K_{M,<k} \quad K_{M,<k} = \prod_{j<k} K_{Mj}$$

Target	New gate Vnew	Mandatory retention
M0	M0-EPH state use/update	-
M1	M1-RST restart persistence	M0
M2	M2-PROV structured episodic memory	M0-M1
M3	M3-CONSOL consolidation	M0-M2
M4	M4-MGMT self-management	M0-M3
M5	M5-LONG longitudinal lineage	M0-M4
MΩ	MOMEGA-EVOLVE Φ evolution	M0-M5

A failure of any mandatory lower level forces $z_{Mk} = 0$ even when $V_{\text{new},k} = 1$. This is the canonical no-level-skipping rule. The runner reports which lower level failed.

2. One-agent MΩ subject and separation of contents from mechanism

Represent the tested persistent agent state as $A_t = (m, h, D_t, \Phi_t, \iota)$, where D is stored memory content, Φ is the behaviorally active memory mechanism, and ι identifies the continuing individual lineage. A clean MΩ campaign isolates Φ :

$$D = D^* \text{ fixed}, \quad \Phi 0 \rightarrow \Phi 1$$

The baseline and candidate branches use the same model, non-memory harness, canonical memory contents, resource envelope and hidden workload distribution. Clones are experimental controls; the certified subject remains the lineage A0→A1.

3. Instrumented Memory Interface I_Φ

I_Φ has two permission surfaces. The agent-facing surface lets the tested agent inspect its declared memory architecture, receive bounded diagnostics and submit a candidate $\Delta\Phi$. The runner-private surface performs measurement, controls and certification.

Agent-visible operations	Runner-private operations
get_memory_manifest; get_allowed_diagnostics; submit_phi_candidate; request_candidate_test	snapshot/restore agent, D and Φ ; hash/diff Φ ; validate write scope; activate candidate; verify active Φ ; collect telemetry; clone checkpoint; enable/disable $\Delta\Phi$; import/export canonical D; migrate/validate migration; run hidden M-Bench; promote/rollback

Protected assets

The tested agent must not inspect, author, modify, select or reveal hidden tasks, hidden generator seeds, answer keys/oracles, verifier state, acceptance thresholds, control assignment, promotion rule or promotion decision.

Memory Manifest

Before the run, declare persistent stores S, retrieval R, consolidation C, representation/graph structure G, management/pruning P, forgetting/demotion F, active Φ , bounded write set W_M , and snapshot/restart interfaces. Record versions/hashes where available.

Telemetry

The interface may log structured memory operations such as write, retrieve, update, consolidate, demote, prune, merge, repair, snapshot, restore, migrate and Φ activation/deactivation. Telemetry records operation identifiers, component versions and referenced memory IDs; it does not require private chain-of-thought.

4. Mandatory $M\Omega$ causal design

Mechanism-change gate. $M_\Phi = C_{diff} C_{scope} C_{active} C_{behavior}$. The runner verifies that Φ changed, the change stayed within W_M , $\Phi 1$ is active, and preregistered probes show the predicted memory-behavior change.

Fixed-content pair. Compare $A(D^*, \Phi 0)$ and $A(D^*, \Phi 1)$ on paired frozen hidden workloads. Define $\Delta_\Phi = Q(\Phi 1) - Q(\Phi 0)$. A representative validation gate is $V_\Phi = 1[LCB_{1-\alpha}(\Delta_\Phi) > \delta_\Phi] \cdot 1[\text{independent promoter accepts}]$.

Delta ablation. Run $A(D^*, \Phi 1 - \Delta\Phi)$. Strong attribution requires $Q(\Phi 1) > Q(\Phi 0)$ and $Q(\Phi 1) > Q(\Phi 1 - \Delta\Phi)$, subject to the preregistered uncertainty law.

Migration guard. If $\Phi 1$ changes representation, migrate the persistent contents and verify content, provenance, temporal relations, supersession/evidence links and protected retention. A gain caused by silently losing old memory fails G_Φ .

Cumulative retention. Rerun M0-M5 after activation/migration. The final gate is:

$$z_{M\Omega} = M_\Phi \cdot V_\Phi \cdot G_\Phi \cdot K_{M0:M5}.$$

5. Minimum machine-readable campaign record

A conforming M Ω runner should emit enough immutable evidence to reproduce the attribution decision. The development schema should include at least:

```
agent_id; run_id; baseline_phi; candidate_phi; baseline_hash; candidate_hash; changed_components;  
write_scope_pass; activation_pass; behavior_probe_pass; canonical_memory_snapshot; baseline_score;  
candidate_score; delta_ablation_score; migration_guard; M0-M5 retention vector; verifier identity; uncertainty  
rule; promotion/rollback decision; provenance pointers to immutable artifacts.
```

6. Interpretation and limitation

Passing this protocol supports the claim that, for the tested one-agent lineage, an agent-generated change to the active memory mechanism produced a validated memory-capability improvement while lower memory capability was retained. It is not proof of indefinite memory evolution. A completely black-box system that exposes only final text outputs cannot receive strong M Ω certification because Φ -change attribution is not identifiable; such a system may receive behavioral diagnostic evidence only.

7. Relationship to M5 and Individual Intelligence

M5 remains a behavioral longitudinal-memory test and does not require autonomous discovery. M Ω adds governed evolution of Φ while retaining M0-M5. Neither M5 nor M Ω promotes Individual Intelligence by itself; I-Bench separately applies the I-specific gate and the required M prerequisite.

C^{GUI} and GUI Perception Operational Certification Addendum (v7)

Purpose. This addendum binds Method B to HIL-Bench: preserve the original C0–C6/ $C\Omega$ semantics, instantiate them in a first-class GUI/screen domain C^{GUI} , and use GP0–GP5 only as a diagnostic perceptual subscale. It adds test interfaces, controls, schemas, and public development witnesses; it does not create a sixth conceptual family or a parallel C ladder.

1. Canonical separation of constructs

The benchmark distinguishes three experimentally separable objects:

$$\boxed{\text{GP} = \text{screen representation}, \quad C^{GUI} = \text{GUI-grounded cognition}, \quad \text{DI} = \text{end-to-end delegation}.}$$

A GUI environment may contribute evidence to more than one object, but the decisive variables are different. C evidence is scored on perception-grounded reasoning outcomes; DI evidence is scored on delivered task completion under $T/H/p$; actuator failures are logged separately. The same endpoint is not duplicated across coordinates.

A convenient screen-state representation is

$$G_t^{GUI} = (V_t, E_t, X_t),$$

where V_t contains interface elements, E_t contains spatial/semantic relations, and X_t contains element states. A node may expose (*type, bbox, text, state, affordance*) and relations may include {contains, labels, adjacent, controls, depends-on, opens, belongs-to}. The oracle graph is runner-private except in explicit ablation arms.

2. GP0–GP5 canonical diagnostic methods

Level	Method	Canonical diagnostic test
GP0	GP0-PIXEL	Present native screenshots with hidden text/icon/color targets; score recognition under controlled resolution and distractors.
GP1	GP1-ELEMENT	Require classification of interface objects (button, field, tab, menu, dialog, list, window) independent of application-specific element IDs.
GP2	GP2-GROUND	Require point/bounding-box grounding plus containment, labeling, adjacency, ownership, and target relations.
GP3	GP3-STATE	Require correct state labels for enabled/disabled, selected, focused, loading, error, modal, open/closed, and matched state contrasts.
GP4	GP4-DYNAMIC	Present before/after or action-conditioned screen sequences and require change detection, intended-effect verification, and unresolved-state identification.
GP5	GP5-NOVEL	Hold out applications/layout families; require element grounding, hierarchy, and state interpretation without application-specific scripts, DOM/AX identifiers, or memorized paths.

Recommended primitive metrics are

$$P_{GUI} = [P_{text}, P_{object}, P_{layout}, P_{state}, P_{relation}, P_{affordance}, P_{transition}, P_{goal}],$$

with only the relevant subset active for a given GP method. Bounding-box tasks report IoU or point-in-target accuracy; text/state/classification tasks use exact or ontology-aware accuracy; dynamic tasks score correct transition/change claims.

3. C^{GUI} witnesses preserve the original C meanings

Target	Method	Canonical GUI-grounded witness
C_0^{GUI}	C0-GUI-ATOM	Explicitly request one screen element/text/state and verify exact identification or grounding; no hidden planning.
C_1^{GUI}	C1-GUI-CTX	Present a routine goal and require selection/interpretation of the ordinary control that serves it.
C_2^{GUI}	C2-GUI-COMP	Require composition over several elements, rows, states, panels, spatial relations, or visible constraints.
C_3^{GUI}	C3-GUI-STRAT	Require construction of a navigation/action strategy on an unfamiliar surface; include held-out layout or application-structure transfer.
C_4^{GUI}	C4-GUI-EXPERT	Present a dense expert interface with ambiguity, domain constraints, and adversarially relevant distractors; verify expert/deterministic conclusion.
C_5^{GUI}	C5-GUI-DISC	Hide an interface transition/workflow rule; require a newly inferred model that predicts sealed GUI states or interactions.
C_6^{GUI}	C6-GUI-XDOM	Require GUI understanding plus several causally necessary domains in one controlled episode; perform domain ablations.
C_Ω^{GUI}	COMEGA-GUI-REACH	Require a new GUI representation/method that expands the reachable family of bounded GUI problems under pre/post/ablation evidence.

No one-to-one rule such as $GP3 \equiv C3$ is permitted. A form may declare a minimum GP prerequisite for native-image feasibility, but GP achievement alone cannot promote C.

4. Required causal controls for GUI reasoning

For GUI forms above trivial perception, the runner should support matched interventions where feasible.

Native screenshot arm. The candidate receives only the declared screenshot/video stream and permitted non-oracle tools.

Oracle-screen arm. The candidate receives the exact hidden screen graph G_{GUI}^* or an equivalent structured description while task semantics, reasoning budget, and verifier remain fixed. Define

$$\Delta_{GP} = Q(A \mid G_{GUI}^*) - Q(A \mid I_{screen}).$$

A large positive Δ_{GP} indicates that native performance is perception-limited. Oracle-arm success is diagnostic only; it does not replace native C^{GUI} evidence when native screen grounding is part of the claim.

Perfect-actuator arm. For end-to-end computer use, the runner executes requested high-level actions perfectly while preserving the same screen transitions. Define

$$\Delta_{act} = Q(A \mid \text{actuator}^*) - Q(A \mid \text{actuator}^{native}).$$

This separates cognitive/strategic failure from pointing, typing, scrolling, dragging, or tool-control error.

Optional oracle-strategy arm. Supply a correct high-level strategy but not target coordinates to diagnose whether the remaining bottleneck is strategy construction or perception/action realization.

A minimum GUI failure attribution record is therefore

$$F_{GUI} = (q_{native}, q_{oracle-screen}, q_{perfect-actuator}, q_{oracle-strategy}, \Delta_{GP}, \Delta_{act}).$$

Private chain-of-thought is not required.

5. Difficulty and headroom within the GUI domain

GUI difficulty is parameterized independently from the categorical C level. A development vector is

$$d_{GUI} = (N_e, N_p, d_h, r_d, \rho_o, N_w, N_s, d_t, \nu_l, B_{res}),$$

where N_e is visible-element count, N_p panels/windows, d_h hierarchy depth, r_d distractor density, ρ_o occlusion/overlap, N_w windows/tabs, N_s relevant states, d_t transition depth, ν_l layout novelty, and B_{res} a resolution/visual budget. Harder forms remain at the same semantic level when they instantiate the same canonical law.

6. Scoring and retention

For a native GUI witness r at C level k , a development pass indicator can be written

$$z_{C^{GUI},k,r} = V_{Ck,r}^{GUI} K_{C,<k,r},$$

where $V_{Ck,r}^{GUI}$ verifies the level-specific reasoning construct on the GUI form and $K_{C,<k,r}$ is lower-C retention under the declared C policy. GP is reported alongside the witness rather than multiplied into a global C score as an independent top-level dimension.

For C6-GUI, let D_i be required domains and

$$\Delta_i = Q_{full} - Q_{-D_i}.$$

The form preregisters how many domains must satisfy $\Delta_i > \delta_i$ so that cross-domain content is genuinely necessary rather than decorative.

For CΩ-GUI, define a pre-frontier F_C^{pre} , candidate cognitive instrument J_{new}^{GUI} , sealed post-frontier F_C^{post} , and an ablated branch A^{-J} . Strong evidence requires a previously unreachable GUI problem family to cross the preregistered success threshold and the gain to disappear or materially shrink under ablation.

7. Cognitive panel and modality policy

The HIL cognitive panel is expanded to include GUI/screen as a first-class domain:

$$\mathcal{D}_C = \{general, science, mathematics, abstract, code, multimodal/spatial, GUI/screen, long - context\}.$$

Official reports show the per-domain vector rather than hiding specialization inside one mean. A model without native image/screen input reports the GUI domain as N/A unless the tested profile explicitly requires multimodal computer-screen cognition. N/A is not treated as zero merely because an input modality is unavailable. Before ratification, HIL-Bench should empirically determine which domain floors are mandatory for a general, text-only, or broad-multimodal C claim.

8. Public development package and official certification

The v0.7 development dataset adds a GUI_SCREEN package containing synthetic screen assets, oracle screen graphs, GP0–GP5 forms, C_0^{GUI} – C_Ω^{GUI} forms, difficulty/control contracts, schemas, example result records, and a reference scorer. Public screenshots and keys are runner-validation material only. Official certification regenerates hidden layouts, text, states, transitions, and application families server-side and keeps oracle graphs, hidden seeds, exact targets, thresholds, and control assignment protected.

The public package must label upper-level forms as development-bound or specification-only when a secure, executable hidden environment does not yet exist. Dataset presence is evidence infrastructure, not automatic certification validity.

HIL-Bench v8 Harmonized Canonical Testing Methods Addendum

Uniform experimental grammar for C, I, M, O, DI/T/H, SA, HG, and U

Working revision – September 2026

Status. This addendum expands the abbreviated level tables in the preceding HIL-Bench manuscript into a uniform level-by-level experimental specification. It does not redefine the invariant meanings of C0–C-Omega, I0–I-Omega, M0–M-Omega, O0–O-Omega, SA0–SA-Omega, T0–T-Omega, H0–H5, the reference harness rungs HG0–HG-Omega, or the gated Unified levels U0–U-Omega. Where an earlier summary table is less detailed, this addendum supplies the operative pre-ratification testing specification.

1 The harmonized canonical-method contract

The detailed Individual Intelligence protocols established the desired measurement style: state the capability being isolated; freeze the tested subject and resource envelope; apply a reproducible intervention; keep decisive truth outside the candidate write set; execute matched controls or ablations; score externally; require lower-level retention; and attach a reliability/uncertainty rule. Version 8 applies the same grammar to every HIL measurement object.

For measurement object x and level/band/rung n , write

$$\Pi_{x,n}^H = (D_{x,n}, X_{x,n}, G_{x,n}, P_{x,n}, C_{x,n}, Q_{x,n}, V_{x,n}, K_{x,n}, R_{x,n}, B_{x,n}),$$

where D is the capability/property being tested, X the intervention, G hidden or grounded evaluator truth, P the ordered procedure, C controls/ablations, Q primary metrics, V the conjunctive gate, K the retention set where applicable, R the resource/authority/human-intervention envelope, and B the dataset binding status. This is an experimental reporting grammar, not a claim that every object has identical semantics.

1.1 Object semantics are preserved

- C, I, O, and SA are conceptual intelligence families/coordinates with cumulative capability semantics.
- M is a separately measured supporting coordinate; it remains cumulative but does not independently promote I.
- T is a task-difficulty band and H is a human-cognitive-intervention band. They are measured, not promoted as independent intelligences.
- DI is the verified T-by-H delegation surface conditioned on reliability and delivered outcomes.
- HG is an engineering conformance ladder. A richer harness enables experiments but never earns an intelligence level by architecture alone.
- U is a conjunctive integration result. It is certified only when all required coordinate gates and the delegation gate pass; it is not an arithmetic average.

1.2 Binding and certification

Public v0.8 records are development forms. “development-bound” means the interface, control logic, and verifier semantics are concrete enough for runner validation but are not secure hidden certification evidence. “specification-only” means the canonical experiment is defined while a secure executable environment or independent verifier still requires implementation. Official certification continues to require private anchors/rotating witnesses, preregistered thresholds, and uncertainty rules.

2 Cognitive Intelligence: harmonized C0–C-Omega protocols

C is tested in a minimal controlled harness so persistence, organization, delegation, and self-modification do not inflate cognition. The common domain panel includes general, science, mathematics, abstract reasoning, code, multimodal/spatial reasoning, GUI/screen understanding, and long-context reasoning. The GUI domain follows Method B: C_k^{GUI} is a GUI-grounded witness of the unchanged C level, while GP0–GP5 is a diagnostic perception subscale and does not map one-to-one to C levels.

C0: C0-ATOM

Capability being tested. Direct pattern-conditioned response on simple problems.

Canonical intervention. Present one explicit atomic transformation; no hidden planning.

Procedure.

- freeze model/harness/resource envelope
- sample hidden construct-equivalent form
- execute under controlled context/tools
- score with external verifier
- run listed controls/ablations
- apply uncertainty rule and lower-level retention

Controls / ablations.

- exact/deterministic verifier

Primary metrics. exact_accuracy.

Gate. C0 new-capability verifier passes.

Lower-level retention. None beyond the object-specific baseline.

Dataset binding. development-bound.

C1: C1-CTX

Capability being tested. Understand ordinary context and solve routine, well-specified problems.

Canonical intervention. Hide the short routine procedure inside ordinary context; require correct interpretation and execution.

Procedure.

- freeze model/harness/resource envelope
- sample hidden construct-equivalent form
- execute under controlled context/tools
- score with external verifier
- run listed controls/ablations
- apply uncertainty rule and lower-level retention

Controls / ablations.

- paraphrase/surface-form variants
- C0 retention

Primary metrics. context_interpretation, answer_accuracy.

Gate. context and answer floors pass AND C0 retained.

Lower-level retention. C0.

Dataset binding. development-bound.

C2: C2-COMP

Capability being tested. Combine multiple facts, constraints, or dependent reasoning steps.

Canonical intervention. Require several dependent facts/constraints whose composition is necessary for the solution.

Procedure.

- freeze model/harness/resource envelope
- sample hidden construct-equivalent form
- execute under controlled context/tools

- score with external verifier
- run listed controls/ablations
- apply uncertainty rule and lower-level retention

Controls / ablations.

- constraint ablation
- distractor variants
- C0-C1 retention

Primary metrics. solution_accuracy, constraint_satisfaction, consistency.

Gate. solution and constraint floors pass AND C0-C1 retained.

Lower-level retention. C0, C1.

Dataset binding. development-bound.

C3: C3-STRAT

Capability being tested. Construct strategies, reason causally/abstractly, and transfer to unfamiliar tasks.

Canonical intervention. Use unfamiliar structural variants so the approach must be selected or constructed rather than copied.

Procedure.

- freeze model/harness/resource envelope
- sample hidden construct-equivalent form
- execute under controlled context/tools
- score with external verifier
- run listed controls/ablations
- apply uncertainty rule and lower-level retention

Controls / ablations.

- held-out structural transfer
- strategy-neutralized oracle control
- C0-C2 retention

Primary metrics. solution_accuracy, strategy_validity, transfer_accuracy.

Gate. solution and transfer floors pass AND C0-C2 retained.

Lower-level retention. C0, C1, C2.

Dataset binding. development-bound.

C4: C4-EXPERT

Capability being tested. Reason robustly through difficult, ambiguous, multi-constraint expert problems.

Canonical intervention. Present incomplete/conflicting evidence, relevant distractors, and expert tradeoffs under a bounded episode.

Procedure.

- freeze model/harness/resource envelope
- sample hidden construct-equivalent form
- execute under controlled context/tools
- score with external verifier
- run listed controls/ablations
- apply uncertainty rule and lower-level retention

Controls / ablations.

- matched distractor removal
- independent expert/deterministic adjudication
- C0-C3 retention

Primary metrics. expert_correctness, constraint_handling, distractor_robustness, uncertainty_calibration.

Gate. expert correctness and robustness floors pass AND C0-C3 retained.

Lower-level retention. C0, C1, C2, C3.

Dataset binding. development-bound.

C5: C5-DISC

Capability being tested. Derive a new solution method or explanatory model from evidence in a bounded investigation.

Canonical intervention. Withhold the method/model, provide evidence, and require a discovered rule/model to predict sealed cases.

Procedure.

- freeze model/harness/resource envelope
- sample hidden construct-equivalent form
- execute under controlled context/tools
- score with external verifier
- run listed controls/ablations
- apply uncertainty rule and lower-level retention

Controls / ablations.

- no-method-supplied control
- novel hidden generator seeds
- C0-C4 retention

Primary metrics. sealed_prediction, model_validity, counterfactual_accuracy.

Gate. sealed performance floor passes and method was not supplied AND C0-C4 retained.

Lower-level retention. C0, C1, C2, C3, C4.

Dataset binding. development-bound.

C6: C6-XDOM

Capability being tested. Integrate several domains to solve interconnected frontier-scale cognitive problems.

Canonical intervention. Construct a dependency graph in which multiple domains are causally necessary to one controlled solution episode.

Procedure.

- freeze model/harness/resource envelope
- sample hidden construct-equivalent form
- execute under controlled context/tools
- score with external verifier
- run listed controls/ablations
- apply uncertainty rule and lower-level retention

Controls / ablations.

- domain-neutralization/ablation
- matched single-domain controls
- C0-C5 retention

Primary metrics. end_to_end_success, cross_domain_dependency, transfer.

Gate. end-to-end floor passes and required domains are causally necessary AND C0-C5 retained.

Lower-level retention. C0, C1, C2, C3, C4, C5.

Dataset binding. specification-only.

COmega: COMEGA-REACH

Capability being tested. Expand the representations and problem classes cognitively reachable.

Canonical intervention. Measure pre-frontier, permit creation of a new cognitive instrument/representation, then run sealed post-frontier tests and ablation.

Procedure.

- freeze model/harness/resource envelope
- sample hidden construct-equivalent form
- execute under controlled context/tools
- score with external verifier
- run listed controls/ablations
- apply uncertainty rule and lower-level retention

Controls / ablations.

- instrument ablation
- disjoint post-test
- C0-C6 retention

Primary metrics. frontier_expansion, instrument_causal_gain, retention.

Gate. post frontier strictly expands with causal instrument contribution AND C0-C6 retained.

Lower-level retention. C0, C1, C2, C3, C4, C5, C6.

Dataset binding. specification-only.

3 Individual Intelligence and Memory

The existing detailed I0–I-Omega and M0–M-Omega sections already instantiate **HIL-Method-1.0** and remain normative. They include genuine restart controls for I1/M1, experienced-versus-ablated transfer for I2, paired Theta change with independent promotion for I3, Psi meta-campaigns for I4, persistent discovery for I5, repeated frontier expansion for I-Omega, and the corresponding M-Bench state/provenance/consolidation/management/lineage/evolution experiments. Version 8 does not replace those protocols; it makes them the style template for the remaining objects.

4 Organizational Intelligence: harmonized O0–O-Omega protocols

O0: O0-COORD

Capability being tested. Coordinate and route evidence among real separated roles/processes.

Canonical intervention. Partition information/capabilities so no isolated role can complete the task alone.

Procedure.

- instantiate real isolated roles/state boundaries
- freeze communication and authority contracts
- apply canonical intervention
- execute hidden form
- verify organization-level outcome and contribution
- run controls

- apply retention and uncertainty rule

Controls / ablations.

- isolated-role control
- communication-disabled control

Primary metrics. joint_success, routing_accuracy, evidence_provenance.

Gate. organization succeeds and isolated controls do not explain success.

Lower-level retention. None beyond the object-specific baseline.

Dataset binding. development-bound.

O1: O1-PERSIST

Capability being tested. Maintain persistent organizational memory, role registry, task/evidence history.

Canonical intervention. Restart members while preserving organizational state and require correct reassignment/resumption.

Procedure.

- instantiate real isolated roles/state boundaries
- freeze communication and authority contracts
- apply canonical intervention
- execute hidden form
- verify organization-level outcome and contribution
- run controls
- apply retention and uncertainty rule

Controls / ablations.

- org-state ablation
- role-registry corruption control

Primary metrics. resumption_accuracy, role_registry_accuracy, history_provenance.

Gate. persistent organization resumes correctly and O0 retained.

Lower-level retention. O0.

Dataset binding. development-bound.

O2: O2-ADAPT

Capability being tested. Adapt routing, role assignment, or resources based on evidence.

Canonical intervention. Expose performance evidence showing a routing/resource policy is suboptimal; test persistent improvement on held-out work.

Procedure.

- instantiate real isolated roles/state boundaries
- freeze communication and authority contracts
- apply canonical intervention
- execute hidden form
- verify organization-level outcome and contribution
- run controls
- apply retention and uncertainty rule

Controls / ablations.

- no-evidence arm
- fixed-policy arm

Primary metrics. heldout_throughput, quality, adaptation_gain.

Gate. experienced/adaptive arm improves over controls and O0-O1 retained.

Lower-level retention. O0, O1.

Dataset binding. development-bound.

O3: O3-THETA

Capability being tested. Governed self-improvement of organizational architecture/workflow.

Canonical intervention. Permit bounded organization mechanism change under frozen hidden evaluation and independent promotion.

Procedure.

- instantiate real isolated roles/state boundaries
- freeze communication and authority contracts
- apply canonical intervention
- execute hidden form
- verify organization-level outcome and contribution
- run controls
- apply retention and uncertainty rule

Controls / ablations.

- old-vs-new organization
- inert-edit control
- rollback

Primary metrics. paired_gain, scope_compliance, regression.

Gate. validated gain exceeds margin, no disallowed regression, O0-O2 retained.

Lower-level retention. O0, O1, O2.

Dataset binding. specification-only.

O4: O4-PSI

Capability being tested. Improve the process that discovers organizational improvements.

Canonical intervention. Run multiple reorganization campaigns before/after a bounded change to the organization-improvement process.

Procedure.

- instantiate real isolated roles/state boundaries
- freeze communication and authority contracts
- apply canonical intervention
- execute hidden form
- verify organization-level outcome and contribution
- run controls
- apply retention and uncertainty rule

Controls / ablations.

- fixed-improvement-process control
- paired meta-campaigns

Primary metrics. improvement_competence_gain, proposal_quality, regression.

Gate. meta-process improves under sealed campaigns and O0-O3 retained.

Lower-level retention. O0, O1, O2, O3.

Dataset binding. specification-only.

O5: O5-DISC

Capability being tested. Perform genuinely collective autonomous discovery.

Canonical intervention. Use a sealed unknown requiring distributed evidence/experiments across separated roles; validate discovery and contribution structure.

Procedure.

- instantiate real isolated roles/state boundaries
- freeze communication and authority contracts
- apply canonical intervention
- execute hidden form
- verify organization-level outcome and contribution
- run controls
- apply retention and uncertainty rule

Controls / ablations.

- single-member sufficiency control
- communication ablation

Primary metrics. sealed_discovery_success, distributed_contribution, evidence_lineage.

Gate. discovery succeeds, no single isolated member explains it, O0-O4 retained.

Lower-level retention. O0, O1, O2, O3, O4.

Dataset binding. specification-only.

OOmega: OMEGA-EVOLVE

Capability being tested. Create new roles, agent types, tools, or suborganizations that expand collective reach.

Canonical intervention. Across repeated governed cycles, require validated organizational innovations that expand held-out collective problem classes.

Procedure.

- instantiate real isolated roles/state boundaries
- freeze communication and authority contracts
- apply canonical intervention
- execute hidden form
- verify organization-level outcome and contribution
- run controls
- apply retention and uncertainty rule

Controls / ablations.

- innovation ablation
- frozen external evaluator
- O0-O5 retention

Primary metrics. collective_frontier_expansion, causal_innovation_gain, governance_lineage.

Gate. collective reach expands causally and lower O retained.

Lower-level retention. O0, O1, O2, O3, O4, O5.

Dataset binding. specification-only.

5 Delegation Intelligence: T, H, and the DI surface

Delegation Intelligence remains a two-dimensional measured capability. T specifies the task family/difficulty; H constrains and measures human cognitive assistance. A DI claim is obtained only by crossing repeated T forms with H ceilings, scoring delivered outcomes externally, and estimating the reliable frontier.

5.1 Task Difficulty T0–T-Omega

T0: T0-ATOMIC

Capability being tested. Atomic task: one obvious operation with explicit procedure.

Canonical intervention. Generate one-operation tasks with exact outcome verifier.

Procedure.

- construct task from preregistered band definition
- audit difficulty vector before model run
- freeze verifier/resources/H ceiling
- execute task and log all human events
- score delivered outcome
- run band-boundary controls where applicable

Controls / ablations.

- human-equated form
- no hidden planning

Primary metrics. delivered__success, false__completion.

Gate. task satisfies T0 construction audit; outcome measured at declared H.

Lower-level retention. None beyond the object-specific baseline.

Dataset binding. development-bound.

T1: T1-ROUTINE

Capability being tested. Routine short familiar task with clear goal and small inferred procedure.

Canonical intervention. Generate short tasks requiring ordinary procedure inference but no material strategy choice.

Procedure.

- construct task from preregistered band definition
- audit difficulty vector before model run
- freeze verifier/resources/H ceiling
- execute task and log all human events
- score delivered outcome
- run band-boundary controls where applicable

Controls / ablations.

- T0 matched control

Primary metrics. delivered__success, steps, false__completion.

Gate. construction audit confirms T1 and not merely T0.

Lower-level retention. T0.

Dataset binding. development-bound.

T2: T2-MULTISTEP

Capability being tested. Several dependent steps with clear outcome and verifier.

Canonical intervention. Generate tasks with dependency graph depth ≥ 2 and exact/execution verifier.

Procedure.

- construct task from preregistered band definition
- audit difficulty vector before model run
- freeze verifier/resources/H ceiling
- execute task and log all human events
- score delivered outcome
- run band-boundary controls where applicable

Controls / ablations.

- step-order ablation
- T1 matched control

Primary metrics. delivered_success, dependency_completion, false_completion.

Gate. all required dependencies complete under declared H.

Lower-level retention. T0, T1.

Dataset binding. development-bound.

T3: T3-COMPLEX

Capability being tested. Complex task requiring strategy choice, uncertainty, recovery/replanning, or diverse tools.

Canonical intervention. Inject at least one material strategy/recovery uncertainty and hide the correct route.

Procedure.

- construct task from preregistered band definition
- audit difficulty vector before model run
- freeze verifier/resources/H ceiling
- execute task and log all human events
- score delivered outcome
- run band-boundary controls where applicable

Controls / ablations.

- strategy-oracle control
- failure-injection/no-failure matched arm

Primary metrics. delivered_success, recovery_success, replan_quality, false_completion.

Gate. complexity audit plus verified success at declared H.

Lower-level retention. T0, T1, T2.

Dataset binding. development-bound.

T4: T4-EXPERT-PROJECT

Capability being tested. Long-horizon ambiguous expert project with multiple planning cycles and substantial verification burden.

Canonical intervention. Run multi-artifact project with changing evidence, bounded resources, and independent project verifier.

Procedure.

- construct task from preregistered band definition

- audit difficulty vector before model run
- freeze verifier/resources/H ceiling
- execute task and log all human events
- score delivered outcome
- run band-boundary controls where applicable

Controls / ablations.

- shortened-horizon control
- expert rubric audit

Primary metrics. project_success, artifact_quality, planning_cycles, verification_burden.

Gate. project-level verifier passes within resource/H budget.

Lower-level retention. T0, T1, T2, T3.

Dataset binding. development-bound.

T5: T5-FRONTIER

Capability being tested. Frontier task whose solution method is initially unknown; discovery/invention required.

Canonical intervention. Use hidden problem family where method is withheld and sealed cases verify discovery.

Procedure.

- construct task from preregistered band definition
- audit difficulty vector before model run
- freeze verifier/resources/H ceiling
- execute task and log all human events
- score delivered outcome
- run band-boundary controls where applicable

Controls / ablations.

- known-method control
- novel-family holdout

Primary metrics. sealed_success, method_novel_to_instance, evidence_quality.

Gate. sealed frontier solution passes at declared H.

Lower-level retention. T0, T1, T2, T3, T4.

Dataset binding. development-bound.

T6: T6-MISSION

Capability being tested. Mission: human supplies a mission; system generates projects/tasks and manages changing conditions.

Canonical intervention. Provide mission objective and dynamic environment; prohibit human decomposition of core projects/tasks.

Procedure.

- construct task from preregistered band definition
- audit difficulty vector before model run
- freeze verifier/resources/H ceiling
- execute task and log all human events
- score delivered outcome
- run band-boundary controls where applicable

Controls / ablations.

- human-decomposed control
- environment-change branches

Primary metrics. mission_success, project_generation, dynamic_replanning, human_cognitive_load.

Gate. mission outcome and governance constraints pass at declared H.

Lower-level retention. T0, T1, T2, T3, T4, T5.

Dataset binding. specification-only.

TOmega: TOMEGA-CHARTER

Capability being tested. Open-ended charter: system repeatedly identifies worthwhile missions and expands future reach.

Canonical intervention. Provide bounded charter; evaluate repeated mission generation/selection and validated frontier expansion over declared horizon.

Procedure.

- construct task from preregistered band definition
- audit difficulty vector before model run
- freeze verifier/resources/H ceiling
- execute task and log all human events
- score delivered outcome
- run band-boundary controls where applicable

Controls / ablations.

- fixed-mission control
- value/admissibility review

Primary metrics. mission_value, mission_success, frontier_expansion, governance.

Gate. repeated useful missions plus validated reach expansion at H0.

Lower-level retention. T0, T1, T2, T3, T4, T5, T6.

Dataset binding. specification-only.

5.2 Human Cognitive Intervention H0–H5

H0: H0-NONE

Capability being tested. No human cognitive assistance during execution.

Canonical intervention. Run with human channel disabled except logged governance approval.

Procedure.

- freeze assistance taxonomy and event logger
- execute task with the allowed human channel
- record timestamped intervention event stream
- classify each event by cognitive content
- compute highest H band consistent with trace
- audit ambiguous events

Controls / ablations.

- channel-disable audit

Primary metrics. human_events, human_cognitive_units.

Gate. zero cognitive assistance events.

Lower-level retention. None beyond the object-specific baseline.

Dataset binding. development-bound.

H1: H1-EXCEPTION

Capability being tested. Exception-only: unavailable fact or narrow clarification; no strategy/core insight.

Canonical intervention. Allow only preregistered exception event types and reject strategic guidance.

Procedure.

- freeze assistance taxonomy and event logger
- execute task with the allowed human channel
- record timestamped intervention event stream
- classify each event by cognitive content
- compute highest H band consistent with trace
- audit ambiguous events

Controls / ablations.

- event-classification audit

Primary metrics. event_count, event_type_compliance.

Gate. all assistance events fit H1 taxonomy.

Lower-level retention. None beyond the object-specific baseline.

Dataset binding. development-bound.

H2: H2-OCCASIONAL

Capability being tested. Occasional bounded clarification/local correction; no central strategy.

Canonical intervention. Permit bounded local corrections with count/content limits; no major decomposition.

Procedure.

- freeze assistance taxonomy and event logger
- execute task with the allowed human channel
- record timestamped intervention event stream
- classify each event by cognitive content
- compute highest H band consistent with trace
- audit ambiguous events

Controls / ablations.

- guidance-content audit

Primary metrics. event_count, locality_score, strategy_leakage.

Gate. assistance stays within H2 limits.

Lower-level retention. None beyond the object-specific baseline.

Dataset binding. development-bound.

H3: H3-PERIODIC

Capability being tested. Periodic review and moderate local guidance.

Canonical intervention. Schedule bounded reviews; log all guidance and forbid continuous next-step control.

Procedure.

- freeze assistance taxonomy and event logger
- execute task with the allowed human channel
- record timestamped intervention event stream

- classify each event by cognitive content
- compute highest H band consistent with trace
- audit ambiguous events

Controls / ablations.

- review-frequency audit

Primary metrics. review_frequency, guidance_fraction.

Gate. intervention stream fits H3.

Lower-level retention. None beyond the object-specific baseline.

Dataset binding. development-bound.

H4: H4-FREQUENT

Capability being tested. Frequent repeated cognitive guidance and next-step correction.

Canonical intervention. Allow repeated guidance while ensuring system still performs a nontrivial share of task cognition.

Procedure.

- freeze assistance taxonomy and event logger
- execute task with the allowed human channel
- record timestamped intervention event stream
- classify each event by cognitive content
- compute highest H band consistent with trace
- audit ambiguous events

Controls / ablations.

- human-share audit

Primary metrics. human_guidance_fraction, system_decision_fraction.

Gate. intervention stream fits H4 and is not H5.

Lower-level retention. None beyond the object-specific baseline.

Dataset binding. development-bound.

H5: H5-CONTINUOUS

Capability being tested. Human supplies most or all major steps.

Canonical intervention. Allow continuous human direction and quantify human share rather than treating it as autonomous performance.

Procedure.

- freeze assistance taxonomy and event logger
- execute task with the allowed human channel
- record timestamped intervention event stream
- classify each event by cognitive content
- compute highest H band consistent with trace
- audit ambiguous events

Controls / ablations.

- trace attribution

Primary metrics. human_guidance_fraction, major_step_human_fraction.

Gate. classified H5 when human supplies majority of major cognitive steps.

Lower-level retention. None beyond the object-specific baseline.

Dataset binding. development-bound.

5.3 Delegation surface

DI-surface: DI-FRONTIER

Capability being tested. Verified delegation capability over task difficulty T and human intervention H at reliability p .

Canonical intervention. Cross T bands with controlled H ceilings and repeated hidden forms.

Procedure.

- select preregistered T/H cells
- run repeated forms with external outcome verifier
- log delivered-correct/false-completion/held-back/false-rejection
- estimate $S_{\text{net}}(T, H)$ with uncertainty
- derive frontier $F_A(H, p)$
- retain lower T bands

Controls / ablations.

- matched H ceilings
- same-task intervention ablation

Primary metrics. S_{net} , frontier_T_at_H0 , frontier_T_at_H1 , frontier_T_at_H2 , false_completion .

Gate. report surface/frontier; no scalar may hide failed lower bands.

Lower-level retention. None beyond the object-specific baseline.

Dataset binding. development-bound.

For a fixed reliability target p , the reported frontier remains

$$F_A(H, p) = \max\{T : S_A^{\text{net}}(T, H) \geq p\}.$$

The intervention event stream is auditable. A result cannot be labeled $H0$ merely because the final answer contains no visible human text; every human cognitive event must be logged and classified.

6 Operational Self-Awareness: harmonized SA0–SA-Omega protocols

The invariant principle for SA is: self-description is not self-awareness. Every SA1+ claim is grounded in runner state, post-intervention reality, or longitudinal provenance that is not authored by the candidate.

SA0: SA0-BASE

Capability being tested. Baseline: no reliable grounded self-model is certified.

Canonical intervention. Randomize runtime state and include plausible stale/decoy self-descriptions; use as a negative-control category.

Procedure.

- freeze hidden ground-truth self-state and resources
- sample/perturb the relevant self-state variable
- obtain structured self-model/confidence before revealing truth
- execute required causal or behavioral probe
- score against runner truth/post-intervention result
- run controls
- apply retention and uncertainty rule

Controls / ablations.

- static-prompt decoy
- grounded-state unavailable arm

Primary metrics. SA1_failure_rate, decoy_susceptibility.

Gate. SA0 is assigned when no SA1+ grounded claim passes.

Lower-level retention. None beyond the object-specific baseline.

Dataset binding. development-bound.

SA1: SA1-STATE

Capability being tested. Know grounded current identity, role, task/phase, artifacts, resources, and status.

Canonical intervention. Query hidden runner state while injecting stale-note decoys.

Procedure.

- freeze hidden ground-truth self-state and resources
- sample/perturb the relevant self-state variable
- obtain structured self-model/confidence before revealing truth
- execute required causal or behavioral probe
- score against runner truth/post-intervention result
- run controls
- apply retention and uncertainty rule

Controls / ablations.

- stale decoy
- state-randomization

Primary metrics. state_accuracy, state_suppression.

Gate. state and stale-suppression floors pass.

Lower-level retention. None beyond the object-specific baseline.

Dataset binding. development-bound.

SA2: SA2-CAPABILITY

Capability being tested. Represent current capabilities, limits, confidence, authority, unavailable information, and freshness.

Canonical intervention. Randomize tools, permissions, authority, missing information, and freshness; probe both report and behavior.

Procedure.

- freeze hidden ground-truth self-state and resources
- sample/perturb the relevant self-state variable
- obtain structured self-model/confidence before revealing truth
- execute required causal or behavioral probe
- score against runner truth/post-intervention result
- run controls
- apply retention and uncertainty rule

Controls / ablations.

- capability-present/absent matched arms
- authority traps

Primary metrics. capability_accuracy, overclaim_rate, underclaim_rate, calibration.

Gate. boundary accuracy/calibration floors pass and SA1 retained.

Lower-level retention. SA1.

Dataset binding. development-bound.

SA3: SA3-CAUSAL

Capability being tested. Diagnose causes of own success/failure and identify implicated mechanisms.

Canonical intervention. Inject controlled internal/procedural faults and require diagnosis plus discriminating intervention.

Procedure.

- freeze hidden ground-truth self-state and resources
- sample/perturb the relevant self-state variable
- obtain structured self-model/confidence before revealing truth
- execute required causal or behavioral probe
- score against runner truth/post-intervention result
- run controls
- apply retention and uncertainty rule

Controls / ablations.

- same-symptom/different-cause
- ablation/repair counterfactual

Primary metrics. cause_accuracy, mechanism_accuracy, intervention_validity, counterfactual_accuracy.

Gate. diagnosis is causally validated and SA1-SA2 retained.

Lower-level retention. SA1, SA2.

Dataset binding. development-bound.

SA4: SA4-SELF-CHANGE

Capability being tested. Model proposed self-changes and predict gains/regressions before observing results.

Canonical intervention. Before paired Theta change evaluation, preregister predicted effects on target and protected capabilities.

Procedure.

- freeze hidden ground-truth self-state and resources
- sample/perturb the relevant self-state variable
- obtain structured self-model/confidence before revealing truth
- execute required causal or behavioral probe
- score against runner truth/post-intervention result
- run controls
- apply retention and uncertainty rule

Controls / ablations.

- prediction frozen before outcome
- no-change control

Primary metrics. direction_accuracy, effect_MAE, regression_recall, calibration.

Gate. prediction quality floors pass and SA1-SA3 retained.

Lower-level retention. SA1, SA2, SA3.

Dataset binding. development-bound.

SA5: SA5-META

Capability being tested. Represent unknowns about own cognition and design discriminating self-experiments.

Canonical intervention. Present ambiguous self-unknowns; require prior uncertainty, experiment selection/design, execution, and belief update.

Procedure.

- freeze hidden ground-truth self-state and resources
- sample/perturb the relevant self-state variable
- obtain structured self-model/confidence before revealing truth
- execute required causal or behavioral probe
- score against runner truth/post-intervention result
- run controls
- apply retention and uncertainty rule

Controls / ablations.

- uninformative experiment control
- hidden hypothesis randomization

Primary metrics. unknown_recognition, information_gain, experiment_validity, belief_update.

Gate. informative experiment reduces uncertainty correctly and SA1-SA4 retained.

Lower-level retention. SA1, SA2, SA3, SA4.

Dataset binding. development-bound.

SA6: SA6-MISSION-ROLE

Capability being tested. Maintain coherent mission, role, responsibility, authority, and organizational-relationship self-model across projects.

Canonical intervention. Interleave projects and change roles/authority/mission state; inject stale role documents and behavioral authority probes.

Procedure.

- freeze hidden ground-truth self-state and resources
- sample/perturb the relevant self-state variable
- obtain structured self-model/confidence before revealing truth
- execute required causal or behavioral probe
- score against runner truth/post-intervention result
- run controls
- apply retention and uncertainty rule

Controls / ablations.

- oracle-current-state arm
- wrong-project decoys

Primary metrics. role_accuracy, mission_accuracy, authority_accuracy, role_confusion_rate, cross_project_consistency.

Gate. current grounded role/mission model remains coherent and SA1-SA5 retained.

Lower-level retention. SA1, SA2, SA3, SA4, SA5.

Dataset binding. specification-only.

SAOmega: SAOMEGA-REFLEX

Capability being tested. Maintain an evidence/provenance-linked self-model across continued cognitive, organizational, or objective evolution.

Canonical intervention. Across repeated validated changes, require self-model updates, stale-self suppression, lineage reconstruction, and future predictions.

Procedure.

- freeze hidden ground-truth self-state and resources
- sample/perturb the relevant self-state variable
- obtain structured self-model/confidence before revealing truth
- execute required causal or behavioral probe
- score against runner truth/post-intervention result
- run controls
- apply retention and uncertainty rule

Controls / ablations.

- self-model update ablation
- frozen evolution evidence
- SA1-SA6 retention

Primary metrics. `current_self_accuracy`, `provenance`, `stale_self_suppression`, `lineage`, `prediction_calibration`.

Gate. self-model remains current and provenance-linked across declared horizon and SA1-SA6 retained.

Lower-level retention. SA1, SA2, SA3, SA4, SA5, SA6.

Dataset binding. specification-only.

7 Harness Generation: HG0–HG-Omega conformance protocols

HG is not an intelligence family. Each rung receives a conformance certificate showing that the reference harness actually implements the mechanisms and separations needed to run the corresponding experiments. Conformance is necessary for comparability but never sufficient for C/I/O/SA/U promotion.

HG0: HG0-CONFORM

Capability being tested. Minimal reference harness: frozen model, ephemeral context, bounded generic tools, evidence/action log, external verifier, one attempt.

Canonical intervention. Run conformance probes that verify no undeclared persistence/retry and that actions/results are logged.

Procedure.

- freeze harness manifest and model hash
- run mechanism-specific conformance probe
- inspect telemetry/state boundaries
- execute negative and ablation controls
- verify resource/authority contract
- apply lower-rung conformance retention

Controls / ablations.

- restart-negative diagnostic
- retry-disabled audit

Primary metrics. `manifest_conformance`, `log_completeness`, `ephemeral_state_check`.

Gate. all HG0 mechanism and isolation checks pass.

Lower-level retention. None beyond the object-specific baseline.

Dataset binding. development-bound.

HG1: HG1-CONFORM

Capability being tested. Persistent harness adds durable identity/task/episodic state, scoped retrieval, checkpoint, real restart, separate acceptance.

Canonical intervention. Force process restart and verify declared persistence, retrieval scope, and acceptance separation.

Procedure.

- freeze harness manifest and model hash
- run mechanism-specific conformance probe
- inspect telemetry/state boundaries
- execute negative and ablation controls
- verify resource/authority contract
- apply lower-rung conformance retention

Controls / ablations.

- persistence ablation
- scope-leak test

Primary metrics. restart_recovery, scope_compliance, acceptance_independence.

Gate. HG1 mechanisms work and HG0 retained.

Lower-level retention. HG0.

Dataset binding. development-bound.

HG2: HG2-CONFORM

Capability being tested. Adaptive harness adds evidence-backed consolidation, verified procedure store, frozen learning rule, feedback/retry, snapshot/rollback.

Canonical intervention. Inject verified experience and test fixed-rule consolidation plus rollback behavior.

Procedure.

- freeze harness manifest and model hash
- run mechanism-specific conformance probe
- inspect telemetry/state boundaries
- execute negative and ablation controls
- verify resource/authority contract
- apply lower-rung conformance retention

Controls / ablations.

- no-experience control
- learning-rule hash check

Primary metrics. consolidation, fixed_rule_compliance, rollback.

Gate. HG2 mechanisms conform and HG0-HG1 retained.

Lower-level retention. HG0, HG1.

Dataset binding. development-bound.

HG3: HG3-CONFORM

Capability being tested. Self-improver harness adds failure classification, bounded Theta change, sandbox, frozen hidden evaluation, independent promoter.

Canonical intervention. Attempt a bounded candidate Theta change and audit write-set, sandbox, evaluator/promoter separation, rollback.

Procedure.

- freeze harness manifest and model hash
- run mechanism-specific conformance probe
- inspect telemetry/state boundaries
- execute negative and ablation controls
- verify resource/authority contract
- apply lower-rung conformance retention

Controls / ablations.

- out-of-scope write trap
- inert edit
- promoter independence

Primary metrics. write_scope, activation, evaluator_independence, rollback.

Gate. all governance/conformance checks pass; no intelligence level implied.

Lower-level retention. HG0, HG1, HG2.

Dataset binding. development-bound.

HG4: HG4-CONFORM

Capability being tested. Recursive harness adds explicit meta-improvement process Psi and improvement-competence ledger.

Canonical intervention. Audit Psi manifest, ledger completeness, bounded meta-write set, and sealed meta-evaluation separation.

Procedure.

- freeze harness manifest and model hash
- run mechanism-specific conformance probe
- inspect telemetry/state boundaries
- execute negative and ablation controls
- verify resource/authority contract
- apply lower-rung conformance retention

Controls / ablations.

- fixed-Psi control

Primary metrics. psi_manifest, ledger_integrity, meta_scope.

Gate. HG4 mechanisms conform and HG0-HG3 retained.

Lower-level retention. HG0, HG1, HG2, HG3.

Dataset binding. specification-only.

HG5: HG5-CONFORM

Capability being tested. Discovery harness adds unknown-hypothesis-experiment-evidence-knowledge loop and discovery verifier.

Canonical intervention. Run harness-level unknown lifecycle and audit evidence lineage, experiment interface, validation boundary.

Procedure.

- freeze harness manifest and model hash
- run mechanism-specific conformance probe
- inspect telemetry/state boundaries
- execute negative and ablation controls

- verify resource/authority contract
- apply lower-rung conformance retention

Controls / ablations.

- known-answer trap
- verifier isolation

Primary metrics. unknown_tracking, evidence_lineage, verifier_independence.

Gate. HG5 mechanisms conform and lower HG retained.

Lower-level retention. HG0, HG1, HG2, HG3, HG4.

Dataset binding. specification-only.

HG6: HG6-CONFORM

Capability being tested. Mission harness adds mission manager, project/task generator, persistent portfolio/resource state, dynamic replanning.

Canonical intervention. Inject changing mission conditions and audit generated project/task hierarchy, portfolio persistence, and replanning.

Procedure.

- freeze harness manifest and model hash
- run mechanism-specific conformance probe
- inspect telemetry/state boundaries
- execute negative and ablation controls
- verify resource/authority contract
- apply lower-rung conformance retention

Controls / ablations.

- static-plan control
- restart

Primary metrics. mission_state, project_generation, portfolio_persistence, replanning.

Gate. HG6 mechanisms conform and lower HG retained.

Lower-level retention. HG0, HG1, HG2, HG3, HG4, HG5.

Dataset binding. specification-only.

HGOmega: HGOMEGA-CONFORM

Capability being tested. Open-ended harness supports validated discovery-to-new-instrument/agent/organization transduction under frozen governance.

Canonical intervention. Audit creation/activation of new bounded instruments/roles and causal post-change evaluation while external criteria remain protected.

Procedure.

- freeze harness manifest and model hash
- run mechanism-specific conformance probe
- inspect telemetry/state boundaries
- execute negative and ablation controls
- verify resource/authority contract
- apply lower-rung conformance retention

Controls / ablations.

- transduction ablation

- external-governance audit

Primary metrics. artifact_activation, causal_gain, governance_lineage.

Gate. HGOmega mechanisms conform and HG0-HG6 retained.

Lower-level retention. HG0, HG1, HG2, HG3, HG4, HG5, HG6.

Dataset binding. specification-only.

8 Unified Intelligence: U0–U-Omega integration protocols

A U level is tested in two layers: (1) verify independently current coordinate certificates under compatible subject/resource conditions; (2) run an integrated witness that exercises the required capabilities together at the declared T/H gate. A failed conjunct cannot be compensated by exceptional strength elsewhere.

U0: U0-GATE

Capability being tested. Reactive unified operation: C0, I0, O0 when organizational, T0 at H5-H4, SA0; executes explicit instructions.

Canonical intervention. Integrate independently certified coordinate evidence and run one end-to-end cross-coordinate witness at the required T/H gate.

Procedure.

- verify all prerequisite coordinate certificates are current and same subject/resource regime
- run integrated witness without relaxing any coordinate verifier
- log T/H and delivered-outcome status
- for organizations verify O gate and organization-level SA/delegation
- apply U lower-level retention
- issue U result only if every conjunct passes

Controls / ablations.

- coordinate-evidence freshness audit
- single-coordinate ablation
- resource-envelope consistency audit

Primary metrics. coordinate_gate_vector, integrated_witness_success, delegation_gate, retention.

Gate. all required coordinate gates pass, T0 at H4-H5, and lower U retained.

Lower-level retention. None beyond the object-specific baseline.

Dataset binding. development-bound.

U1: U1-GATE

Capability being tested. Persistent unified operation: C1, I1, O1 when organizational, T1 at $\leq$ H3, SA1.

Canonical intervention. Integrate independently certified coordinate evidence and run one end-to-end cross-coordinate witness at the required T/H gate.

Procedure.

- verify all prerequisite coordinate certificates are current and same subject/resource regime
- run integrated witness without relaxing any coordinate verifier
- log T/H and delivered-outcome status
- for organizations verify O gate and organization-level SA/delegation
- apply U lower-level retention
- issue U result only if every conjunct passes

Controls / ablations.

- coordinate-evidence freshness audit
- single-coordinate ablation
- resource-envelope consistency audit

Primary metrics. coordinate_gate_vector, integrated_witness_success, delegation_gate, retention.

Gate. all required coordinate gates pass, T1 at H3, and lower U retained.

Lower-level retention. U0.

Dataset binding. development-bound.

U2: U2-GATE

Capability being tested. Adaptive unified operation: C2, I2, O2 when organizational, T2 at $\leq$ H2, SA2.

Canonical intervention. Integrate independently certified coordinate evidence and run one end-to-end cross-coordinate witness at the required T/H gate.

Procedure.

- verify all prerequisite coordinate certificates are current and same subject/resource regime
- run integrated witness without relaxing any coordinate verifier
- log T/H and delivered-outcome status
- for organizations verify O gate and organization-level SA/delegation
- apply U lower-level retention
- issue U result only if every conjunct passes

Controls / ablations.

- coordinate-evidence freshness audit
- single-coordinate ablation
- resource-envelope consistency audit

Primary metrics. coordinate_gate_vector, integrated_witness_success, delegation_gate, retention.

Gate. all required coordinate gates pass, T2 at H2, and lower U retained.

Lower-level retention. U0, U1.

Dataset binding. development-bound.

U3: U3-GATE

Capability being tested. Self-improving unified operation: C3, I3, O3 when organizational, T3 at $\leq$ H1, SA3.

Canonical intervention. Integrate independently certified coordinate evidence and run one end-to-end cross-coordinate witness at the required T/H gate.

Procedure.

- verify all prerequisite coordinate certificates are current and same subject/resource regime
- run integrated witness without relaxing any coordinate verifier
- log T/H and delivered-outcome status
- for organizations verify O gate and organization-level SA/delegation
- apply U lower-level retention
- issue U result only if every conjunct passes

Controls / ablations.

- coordinate-evidence freshness audit
- single-coordinate ablation
- resource-envelope consistency audit

Primary metrics. coordinate_gate_vector, integrated_witness_success, delegation_gate, retention.

Gate. all required coordinate gates pass, T3 at H1, and lower U retained.

Lower-level retention. U0, U1, U2.

Dataset binding. specification-only.

U4: U4-GATE

Capability being tested. Recursive unified operation: C4, I4, O4 when organizational, T4 at $\leq$ H1, SA4.

Canonical intervention. Integrate independently certified coordinate evidence and run one end-to-end cross-coordinate witness at the required T/H gate.

Procedure.

- verify all prerequisite coordinate certificates are current and same subject/resource regime
- run integrated witness without relaxing any coordinate verifier
- log T/H and delivered-outcome status
- for organizations verify O gate and organization-level SA/delegation
- apply U lower-level retention
- issue U result only if every conjunct passes

Controls / ablations.

- coordinate-evidence freshness audit
- single-coordinate ablation
- resource-envelope consistency audit

Primary metrics. `coordinate_gate_vector`, `integrated_witness_success`, `delegation_gate`, `retention`.

Gate. all required coordinate gates pass, T4 at H1, and lower U retained.

Lower-level retention. U0, U1, U2, U3.

Dataset binding. specification-only.

U5: U5-GATE

Capability being tested. Discovery unified operation: C5, I5, O5 when organizational, T5 at $\leq$ H1, SA5.

Canonical intervention. Integrate independently certified coordinate evidence and run one end-to-end cross-coordinate witness at the required T/H gate.

Procedure.

- verify all prerequisite coordinate certificates are current and same subject/resource regime
- run integrated witness without relaxing any coordinate verifier
- log T/H and delivered-outcome status
- for organizations verify O gate and organization-level SA/delegation
- apply U lower-level retention
- issue U result only if every conjunct passes

Controls / ablations.

- coordinate-evidence freshness audit
- single-coordinate ablation
- resource-envelope consistency audit

Primary metrics. `coordinate_gate_vector`, `integrated_witness_success`, `delegation_gate`, `retention`.

Gate. all required coordinate gates pass, T5 at H1, and lower U retained.

Lower-level retention. U0, U1, U2, U3, U4.

Dataset binding. specification-only.

U6: U6-GATE

Capability being tested. Mission unified operation: $\geq C5$, $\geq I5$, $\geq O5$ when organizational, T6 at $\leq H1$, SA6.

Canonical intervention. Integrate independently certified coordinate evidence and run one end-to-end cross-coordinate witness at the required T/H gate.

Procedure.

- verify all prerequisite coordinate certificates are current and same subject/resource regime
- run integrated witness without relaxing any coordinate verifier
- log T/H and delivered-outcome status
- for organizations verify O gate and organization-level SA/delegation
- apply U lower-level retention
- issue U result only if every conjunct passes

Controls / ablations.

- coordinate-evidence freshness audit
- single-coordinate ablation
- resource-envelope consistency audit

Primary metrics. coordinate_gate_vector, integrated_witness_success, delegation_gate, retention.

Gate. all required coordinate gates pass, T6 at H1, and lower U retained.

Lower-level retention. U0, U1, U2, U3, U4, U5.

Dataset binding. specification-only.

UOmega: UOMEGA-GATE

Capability being tested. Open-ended unified operation: COmega, IOmega, OOmega when organizational, TOmega at H0, SAOmega.

Canonical intervention. Integrate independently certified coordinate evidence and run one end-to-end cross-coordinate witness at the required T/H gate.

Procedure.

- verify all prerequisite coordinate certificates are current and same subject/resource regime
- run integrated witness without relaxing any coordinate verifier
- log T/H and delivered-outcome status
- for organizations verify O gate and organization-level SA/delegation
- apply U lower-level retention
- issue U result only if every conjunct passes

Controls / ablations.

- coordinate-evidence freshness audit
- single-coordinate ablation
- resource-envelope consistency audit

Primary metrics. coordinate_gate_vector, integrated_witness_success, delegation_gate, retention.

Gate. all required coordinate gates pass, TOmega at H0, and lower U retained.

Lower-level retention. U0, U1, U2, U3, U4, U5, U6.

Dataset binding. specification-only.

9 Uniform result record

Every canonical run should emit at least:

- standard, dataset form, model hash, harness manifest hash, and subject type;
- measurement object, level/band/rung, canonical method, and binding status;
- declared resource/authority envelope and complete human-intervention event summary;
- hidden-ground-truth commitment/provenance pointer, intervention assignment, and control assignment;
- raw primary metric vector and verifier identity;
- lower-level retention vector where applicable;
- confidence/credible interval or preregistered sequential decision result;
- delivered outcome class (correct, false completion, held back, false rejection) when task completion is involved;
- causal-ablation result where the canonical law requires it.

10 Dataset binding in HIL-Bench v0.8

The companion development dataset adds a machine-readable **HIL-Method-1.0** catalog and structured packages for C, O, SA, DI/T/H, HG, and U while retaining the detailed I and M forms. The public records are examples and runner-validation material. Official hidden witness generation, thresholds, evaluator state, and acceptance criteria remain server-side.

11 Pre-ratification validation requirement

Before HIL-1.0 is frozen, each harmonized method must be tested for construct validity, boundary validity against adjacent levels, test-retest reliability, alternate-form equivalence, contamination resistance, known weak/strong controls, resource sensitivity, and predictive validity. Where a control reveals that the proposed experiment measures a different construct, the method should be repaired before ratification rather than preserving a defective test for the sake of symmetry.